ONCE WE WERE BROTHERS: FORCED MIGRATION, AGRICULTURAL MISMATCH, AND ELECTORAL BEHAVIOR; EVIDENCE FROM THE 1923 LAUSANNE POPULATION EXCHANGE

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ABSTRACT

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Keywords: forced migration, economic disruption, agricultural production, electoral behavior, Lausanne Population Exchange

The 1923 Lausanne population exchange was the first internationally ratified, state-mandated forced migration in the modern era. Despite extensive historical scholarship, its economic and political consequences on the Turkish side remain largely unexamined using fine-grained quantitative data. This thesis constructs a novel dataset covering 51,297 Turkish neighborhoods, classifying as Two-way Exchange, One-way Orthodox Departure, One-way Muslim Exchangee Arrival, or Control. It examines agricultural production and electoral behavior and tests whether agricultural outcomes mediate the exchange's political legacy. Treatment effects are estimated primarily through entropy balancing, with coarsened exact matching and ordinary least squares as robustness checks. A difference-in-differences design compares Ottoman-era agricultural statistics from 1909–1914 with Republican-era production from 1936–1946. Electoral outcomes are analyzed for the 1991 and 2023 elections, followed by causal mediation analysis. Two-way Exchange group exhibit higher olive and tobacco production than Control, contradicting the settler-disruption hypothesis but supporting a mechanism of inherited agricultural capital. The positive association with grape production substantially predates the exchange. Only One-way Muslim Arrival settlements display the negative or null production effects initially predicted. Politically, Two-way Exchange settlements consistently show greater opposition support, lower incumbent support, and lower turnout across both elections. Agricultural production explains only 9-13 percent of this electoral association, suggesting that most of the political legacy operates through other channels. The findings establish that a systematic, persistent electoral difference exists between historically exposed and comparable non-exposed neighborhoods, and that this difference is not primarily explained by the agricultural production channel examined here.

ÖZET

BIR ZAMANLAR KARDEŞTIK: ZORUNLU GÖÇ, TARIMSAL UYUMSUZLUK VE SEÇMEN DAVRANIŞI; 1923 LOZAN NÜFUS MÜBADELESI'NDEN KANITLAR

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Anahtar Kelimeler: zorunlu göç, ekonomik bozulma, tarımsal üretim, seçmen davranışı, Lozan Nüfus Mübadelesi

1923 Lozan nüfus mübadelesi, modern dönemde uluslararası düzeyde onaylanmış ve devlet tarafından zorunlu kılınmış ilk zorunlu göç örneğidir. Kapsamlı tarihsel literatüre rağmen, mübadelenin Türkiye tarafındaki ekonomik ve siyasi sonuçları, ayrıntılı nicel veriler kullanılarak büyük ölçüde incelenmemiştir. Bu tez, Türkiye genelinde 51.297 mahalle ve köyü kapsayan özgün bir veri seti oluşturmakta ve bu yerleşimleri Çift Yönlü Mübadele, Tek Yönlü Ortodoks Ayrılışı, Tek Yönlü Müslüman Mübadil Gelişi veya Kontrol olarak sınıflandırmaktadır. Çalışma, tarımsal üretim ile seçim davranışını incelemekte ve tarımsal sonuçların mübadelenin siyasi mirasına aracılık edip etmediğini test etmektedir. Göç etkileri öncelikle entropi balancing yöntemiyle tahmin edilmekte; sağlamlık kontrolleri olarak coarsened exact matching ve causal mediation yöntemleri kullanılmaktadır. Farkların farkı tasarımı, 1909–1914 dönemine ait Osmanlı tarım istatistiklerini 1936–1946 dönemine ait Cumhuriyet dönemi üretim verileriyle karşılaştırmaktadır. Seçim sonuçları 1991 ve 2023 seçimleri için analiz edilmekte, ardından nedensel aracılık analizi uygulanmaktadır. Çift Yönlü Mübadele grubu, Kontrol grubuna kıyasla daha yüksek zeytin ve tütün üretimi sergilemektedir. Bu bulgu, yerleşimci kaynaklı bozulma hipoteziyle çelişmekte, ancak devralınan tarımsal sermaye mekanizmasını desteklemektedir. Üzüm üretimiyle olan pozitif ilişkinin ise büyük ölçüde mübadele öncesine dayandığı görülmektedir. Yalnızca Tek Yönlü Müslüman Mübadil Gelişi yerleşimleri, başlangıçta öngörülen negatif veya sıfır üretim etkilerini göstermektedir. Siyasi açıdan, Çift Yönlü Mübadele yerleşimleri her iki seçimde de daha yüksek muhalefet desteği, daha düşük iktidar desteği ve daha düşük seçime katılım oranı sergilemektedir. Tarımsal üretim, bu seçimsel ilişkinin yalnızca yüzde 9–13'ünü açıklamakta; bu da siyasi mirasın büyük bölümünün başka kanallar üzerinden işlediğine işaret etmektedir. Bulgular, tarihsel olarak mübadeleye maruz kalmış mahalleler ile karşılaştırılabilir maruz kalmamış mahalleler arasında sistematik ve kalıcı bir seçmen davranışı farkının var olduğunu, ve bu farkın bu tezde incelenen tarımsal üretim kanalıyla birincil olarak açıklanamadığını ortaya koymaktadır.

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1. INTRODUCTION

Forced migration is an enduring global phenomenon that still shapes our world. The displacement of people is a defining feature of the globalized world on a large scale. According to the UNHCR (United Nations High Commissioner for Refugees) (2026), at the end of 2025, 117.8 million people had been forcibly displaced. While 58% consists of internally displaced people, 41.6 million are recognized refugees by the UNHCR.

According to the Groundswell (2018) report, by 2050, a projected 143 million people might be displaced due to the severe effects of ongoing climate change in Latin America, South Asia, and Sub-Saharan Africa. Even though the report estimates these to be internally displaced, it suggests that forced migration is a persistent pattern and likely to increase in scale in the future.

These migratory movements have dramatic consequences for the displaced and the host states, both in terms of economic and political transformations. Forced migration and population exchange are among the most disruptive state interventions in economic and political life. Previous literature on political economy has suggested that other exogenous shocks, similar to forced migration, may generate long-term political legacies (Acemoglu & Robinson, 2013; Grosjean, 2011; Popescu & Popa, 2022; Vogler, 2019). The labor market shocks and shifts in electoral behavior are among the most studied of these transformations. Recent cases in Syria and Ukraine illustrate the drastic impact of cross-border migrations. The economic impact of such migrations was studied particularly for the Syrian refugee crisis, where forced arrivals were shown to disrupt native labor market outcomes, creating a supply shock to the informal labor market (Del Carpio & Wagner, 2015), with corresponding declines in native employment (Ceritoglu et al., 2017). The forced migration also have persistent effects on electoral behaviour and political preferences, which was demonstrated by studies on Ukrainian migratory waves to Poland (Mykhailyshyna & Zuchowski, 2025). This body of work focuses on the host population's response

to arriving migrants. On the other hand, a different strand of the literature traces the long-run economic trajectory of the displaced population (Becker et al., 2020; Bharadwaj et al., 2008; Murard & Sakalli, 2020), while a smaller body of work documents comparable political legacies, such as multi-generational party identification among descendants of displaced populations (C. E. Arbatli & Gomtsyan, 2019). These studies are largely historical and conducted in natural experiment settings.

However, the long-run legacies of the Syrian or Ukrainian cases are yet to be observed, given how recent and ongoing they are. Although these cases illustrate the short-term impact of forced migration, not enough time has passed to observe their long-term economic and political consequences. The Syrian and Ukrainian cases are further confounded by still-active conditions such as ongoing conflict and evolving policy responses, making it difficult to isolate the true effect of forced migration. This raises the central question of this study: does forced displacement leave a durable economic and political legacy on the displaced population itself, and if so, through what mechanism? Answering this question requires a case in which sufficient time has elapsed to observe multi-generational outcomes, and in which the shock can plausibly be treated as independent of local conditions, providing a quasi-experimental setting.

The 1923 Lausanne Convention mandating the exchange of Muslim and Greek Orthodox populations, resettling approximately 395,000 Muslims from Greece to Turkey (Turkiye Buyuk Millet Meclisi, 1932), together with the broader waves of Muslim migration from Rumelia to Anatolia, offer a setting that plausibly satisfies both conditions. Particularly, the Lausanne Convention was the first internationally ratified compulsory population exchange, overseen by the League of Nations (Hirschon, 2003). The compulsory nature of the treaty and the commissions responsible for implementing the exchange in both Greece and Turkey substantially reduced the scope for self-selection relative to voluntary migration, since the decision to relocate and the broad destination were chosen administratively rather than by the migrants themselves. Nevertheless, settlement locations remained shaped by state administrative priorities, logistical considerations, and the availability of housing and land. Consequently, the resulting treatment should be understood as quasi-experimental rather than fully exogenous. While the exchange has been extensively studied with respect to the Greek Orthodox population that relocated to Greece, the Muslim population that resettled in Turkey remains comparatively understudied (Hirschon, 2003; Yıldırım, 2023a). This is particularly true of the exchange's political consequences: the electoral impact of the exchange on Turkey has never been studied as a distinct topic in its own right (Yıldırım, 2023a).

This thesis contributes to the understudied Turkish side of the population exchange and its unstudied political consequences by examining these consequences through an economic lens. This project conducts three analyses, first focusing on estimated logged total agricultural production, second analyzes the electoral behaviour and political preferences by measuring party/alliance vote shares, and finally implementing causal mediation analysis to decompose the effect of agricultural production on electoral behaviour. In addition, a dataset on the ethnic composition of settlements (Nişanyan, 2025) identifies localities associated with both the in-migration and out-migration of ethnic populations, enabling separate comparisons of settlements experiencing refugee arrival, Orthodox departure, or both. In contrast to previous literature, which typically examines either economic or political outcomes in isolation, this thesis tests whether the population exchange disrupted the local agricultural production, and whether it statistically explains the observed association between historical exposure to the population exchange and electoral outcomes. However, due to historical data limitations, this thesis cannot directly identify the underlying mechanism driving electoral outcomes. Instead, it examines agricultural production as one candidate channel through which the economic consequences of the population exchange may be associated with its long-run political outcomes. Furthermore, previous analyses (Kontogiorgi, 2006; Murard, 2023; Murard & Sakalli, 2020) have been conducted at the municipality or district level, whereas this thesis extends the existing literature by integrating neighborhood-level (*mahalle/köy*) settlement and electoral data with spatially reconstructed agricultural estimates. Accordingly, this project examines whether neighborhoods (*mahalle/köy*) historically exposed to *mubadil* settlement exhibit systematically different agricultural production and electoral behaviour relative to otherwise comparable localities.

This thesis documents robust reduced-form patterns and tests one candidate mediating channel rather than claiming full mechanism identification. The agricultural analysis suggests a positive effect on crops that refugees has expertise on, and a left-leaning voting behaviour for 1991 and 2023 elections with lower turnout rate on average.

The remainder of this thesis continues with Chapter 2, providing the historical background to contextualize the case of the Lausanne Population Exchange within broader migration waves, clarifying terminology, and the settlement process following the exchange. Chapter 3 first reviews the literature on forced migration, economic disruption, and the relationship between economic conditions and political outcomes, while also presenting findings specific to Lausanne Population Exchange. It then develops the theoretical framework and forms three testable hypotheses that

structure the empirical analysis: H1 on agricultural production, H2 on electoral behavior, and H3 on the mediating role of agriculture between the two. Chapter 4 describes the data collection and construction of the neighborhood-level dataset, including treatment and control variables from the Ottoman and Republican eras. Chapter 5 outlines the empirical methodology detailing the Entropy Balancing, Coarsened Exact Matching, Difference-in-differences, and Formal Mediation Analysis as an exploratory mechanism analysis. Chapter 6 presents the findings and regression results. Chapter 7 concludes this project by synthesizing these findings and discussing their theoretical and methodological contributions, acknowledging its limitations and outlining directions for future research.

2. HISTORICAL BACKGROUND

2.1 Terminology: *Muhacir* Vs. *Mübadil*

Muhacir, a word from an Arabic origin (*muhājir*), first used to refer to the followers of Prophet Muhammad to Medina, later became a generic term for any Muslim who was forced to migrate from a homeland once under Muslim rule, to another territory ruled by a Muslim sovereign. The term *Muhacir* consequently means "Muslim refugee". *Mübadil*, comes also from Arabic, from the root *bedel* meaning "cost". So *mübadil* corresponds to "the one whose cost has been paid" or "the one who has been exchanged".

These terms and their definition are crucial to distinguish, since in light of the migratory waves from Rumelia to Anatolia, the scope of the terms changes drastically. Even though both are used interchangeably, this thesis will use *mübadil* (exchangee) solely for the exchanged people mandated by the 1923 Lausanne Convention, and *muhacir* (refugee) for any Muslim forcibly migrated to Anatolia. Article 3 of the Convention gave retroactive legal effect to the exchange, classifying all Greeks and Muslims who had already migrated since 18 October 1912 (the beginning of the Balkan Wars) as exchanged persons for the treaty, even though their actual displacement had occurred through war and conflict rather than through the organized exchange process itself (Erol, 2023a, 2023b; Republic of Türkiye Ministry of Foreign Affairs, n.d.). Hence, the term exchangee encompasses all Muslims legally classified as exchanged persons under Article 3. This thesis uses the term *Lozan mübadili* (Lausanne exchangee) to refer to the subset of this legal category who were physically relocated through the organized exchange process itself, between 1923 and 1924. This thesis also focuses on Balkan-origin muhacirs; the term *Rumeli muhaciri* (Rumelia refugee) is used where this distinction matters, to differentiate this popu-

lation from Muslim migrants of Caucasian or other origin who fall under the same broad legal and historical category.

The Lausanne Convention was not an isolated event, but rather the legal formalization of the final major phase of a broader migratory process throughout the late Ottoman period. During the 18th and 19th centuries, there were large-scale Muslim migratory movements to Anatolia from the peripheral lost territories of the Ottoman Empire. Between 1783 and 1913, approximately 5 to 7 million refugees migrated to Ottoman lands from Crimea, Caucasus, Balkans, and from Crete (İnalcık & Quataert, 1994). Particularly after the Russo-Ottoman War (*93 Harbi*) 1877-1878, 2 million Circassians, 1.8 million Tatars arrived in Anatolia. Moreover, another wave occurred during the Balkan Wars, 1912-1913. This is the same date from which Article 3 of the Lausanne Convention granted retroactive legal recognition to displaced persons, discussed above. This migration wave formed part of the population later retroactively classified as exchanged persons under Article 3 of the Lausanne Convention.

2.2 The Lausanne Exchange

The Convention Concerning the Exchange of Greek and Turkish Populations was signed on 30 January 1923, one of several instruments negotiated at the Lausanne Conference and later grouped with the main Treaty of Lausanne, itself signed on 24 July 1923. This convention mandated the compulsory exchange of Orthodox and Muslim populations between Turkey and Greece. The populations were distinguished based on religion, following the logic of the earlier Ottoman *millet* system, which organized subjects by religious community rather than ethnicity or language.

This convention was the first internationally ratified compulsory population exchange, overseen by the League of Nations through its High Commissioner for Refugees, Fridtjof Nansen (Hirschon, 2003). The population exchange stipulated by the Lausanne Treaty, involving Orthodox and Muslim communities, took place over two years between 1923 and 1924. Certain populations were exempted from the exchange: the Orthodox population living in Istanbul, and Muslims living in Western Thrace. Article 3 retroactively classified every displaced person as exchanged from the 18th of October 1912.

In total, approximately 1.2 million Orthodox were forcibly displaced from Anatolia between 1923 and 1924, reflecting both the organized population exchange and displacement associated with the Turkish War of Independence. Overall, approximately 395,000 Muslims from Greece were displaced to Anatolia (Hirschon, 2003). Both governments established a Mixed Commission and pledged that the properties and estates of the relocated populations would be restored upon arrival in their new destinations (Arı, 1995). However, subsequent scholarship suggests that both governments prioritized the rapid implementation of the population exchange without adequately documenting refugee information or property details. While the Convention established the legal and diplomatic framework for the exchange, its implementation on the ground was carried out through separate administrative institutions in each state.

2.3 Settlement Policy And Its Implementation

The Turkish state established a ministry to oversee the settlement of incoming *Lausanne exchangees*: Ministry of Exchange, Reconstruction and Settlement (Mübadele, İmar ve İskan Vekaleti) established in October 1923. This was a short-lived ministry, dissolved in December 1924 and integrated into the Ministry of Interior.

Initially, before the ministry meetings, a special commission in the parliament was gathered to discuss the settlement of incoming Muslim refugees. They decided that refugees from Kavala, Drama, and the Aegean Islands, many of whom were tobacco farmers, would be directed to Tekirdağ and the Aegean coast (Arı, 1995). Those arriving from Salonika were designated for settlement along the Eastern Black Sea region, especially the Samsun coastline, which was considered appropriate for agricultural continuation. The special commission separated Anatolia into ten different regions to receive incoming refugees (*Lausanne exchangees*).

Table 2.1 shows the ten regions that were initially designated for the settlement of incoming refugees.

Table 2.1 The Ten Designated Settlement Zones (*iskân mıntıkaları*)

Zone	**Provinces**
I	Sinop, Samsun, Ordu, Giresun, Trabzon, Gümüşhane, Amasya, Tokat, Çorum
II	Edirne, Tekirdağ, Gelibolu, Kırklareli, Çanakkale
III	Balıkesir
IV	İzmir, Manisa, Aydın, Muğla, Afyon
V	Bursa
VI	İstanbul, Çatalca, Zonguldak
VII	İzmit, Bolu, Bilecik, Eskişehir, Kütahya
VIII	Antalya, Isparta, Burdur
IX	Konya, Niğde, Kayseri, Aksaray, Kırşehir
X	Adana, Mersin, Silifke, Kozan, Gaziantep, Kahramanmaraş

Notes: Arı (1995)

After the ministry was established, its own decree substantially revised this initial distribution, as documented in the finalized regional and occupational breakdown shown in Table 2.2. The ministry also sought information from provinces regarding local crop types, the number of existing buildings, the availability of vacant ones, and the extent of uncultivated land (Arı, 1995). However, the reliability of these surveys was highly questionable, as many of the reported figures were likely exaggerated (Arı, 1995).

The incoming refugees (*Lausanne exchangees*) were predominantly small farmers (*rençber*), growing olive, grape/wine, and tobacco. From a demographic standpoint, the commission classified muhacirs into four occupational categories: tobacco farmers, grape and vine farmers, olive farmers, and general farmers (*rençber*) (Turkiye Buyuk Millet Meclisi, 1932). Their objective was to settle these refugees (*Lausanne exchangees*) in places suitable for their agricultural background. Table 2.2 shows refugee farmers by their agricultural expertise and the ministry's planned settlement.

Table 2.2 Planned Regional Distribution of Incoming Exchangees by Occupation and Destination

Region of Origin	**Tobacco**	**Farmer**	**Vine/Olive**	**Total**	**Destination**
Part of Drama and most of Kavala	30,000	0	0	30,000	Samsun and vicinity
Population of the Serres (Serez) district	20,000	15,000	5,000	40,000	Adana and vicinity
Kozani, Grevena, Neveska, and Kastoria	2,500	15,000	5,000	22,500	Malatya and vicinity
Kailar, Verria, Vodena, Katerini, Elassona, Langadas, Demir Hisar, remaining villages of Gevgelija, Yenice-i Vardar, Karacaabat	3,000	25,000	15,000	43,000	Amasya, Tokat, Sivas
Olive-growers of Drama and Kavala, and the population of Thessaloniki (Selânik)	4,000	20,000	40,000	64,000	Manisa, İzmir, Menteşe, Denizli and vicinity
Kassandra, Polygyros, Sarışaban, Avrethisar, Nevrokop	20,000	55,000	15,000	90,000	Çatalca, Tekirdağ, Karaman districts, Niğde and vicinity
Preveza and Ioannina region	15,000	40,000	0	55,000	Antalya, Silifke and vicinity
Lesbos, Crete, and other islands	0	30,000	20,000	50,000	Ayvalık, Edremit, Mersin and vicinity
Total	**95,000**	**200,000**	**100,000**	**395,000**	

Notes: İskân Tarihçesi (1932), table appended to p. 17 (Turkiye Buyuk Millet Meclisi, 1932).

Despite these efforts, both the ministry and the government lacked the necessary resources and administrative capacity to conduct such investigations efficiently. Furthermore, both the Greek and Turkish governments were eager to complete the exchange as swiftly as possible. However, it was reported that, in practice, these official instructions were not properly followed and were largely disregarded (Arı, 1995). Due to these temporal and technical constraints, settlement locations for the Muslim refugees (*Lausanne exchangees*) were often chosen hastily (Arı, 1995; Yildirim, 2007).

Moreover, the ministry decided to settle refugees predominantly in areas vacated by the departing Orthodox population. The issue of abandoned properties, lands, and

settlements thus became central to the resettlement policy. Member of Parliament Tunalı Hilmi proposed that the incoming refugees could serve as a labor force to rebuild these ruined towns and villages (Arı, 1995; Turkiye Buyuk Millet Meclisi, 1932). However, contemporary newspapers reported that much of the abandoned property and land had already been seized and exploited by residents without legal authorization. This situation further complicated the government's ability to properly resettle muhacirs in those areas. Consequently, rather than identifying regions suitable for agricultural production, the ministry prioritized the reconstruction of the deserted Orthodox villages and neighborhoods (Arı, 1995; Turkiye Buyuk Millet Meclisi, 1932). The primary objective was not necessarily to make immediate use of the fertile land and estates left behind, but rather to revive settlements that had been devastated, either through the departure of the Orthodox population or during the War of Independence.

For logistical reasons, and because much of the relocation was expected to occur during winter, priority was given to areas with easy access to ports, roads, and railways (Arı, 1995; Turkiye Buyuk Millet Meclisi, 1932; Yildirim, 2007). As a result, coastal districts were predominantly selected. Importantly, these considerations suggest that settlement locations were influenced primarily by administrative and logistical factors rather than by the observable characteristics of individual refugees.

The media of the time voiced significant criticism, arguing that government officials failed to respond adequately to the needs of either the local population or the refugees, and that the resettlement process appeared arbitrary and poorly managed (Arı, 1995; Turkiye Buyuk Millet Meclisi, 1932; Yildirim, 2007). Both family accounts and historical records indicate that some muhacirs remigrated domestically after finding their initial settlement unsuitable.

Both governments established a mixed commission and pledged that the properties and estates of the relocated populations would be restored upon arrival in their new destinations (Arı, 1995; Turkiye Buyuk Millet Meclisi, 1932). However, it appears that both sides hastened the implementation of the population exchange without adequately documenting refugee information or property details. Moreover, Greek nationalist groups and segments of the local population were sometimes reported to have forced Muslim residents to leave before official records could be completed. As a result, many refugees were compelled to abandon the wealth they had accumulated over generations or to sell their properties to locals at heavily undervalued prices. Reports indicate that in some cases, the restoration of property rights for refugees took as long as ten years after they arrived in Turkey.

It is estimated that approximately 395,000 Muslims were resettled in Turkey during this exchange (Turkiye Buyuk Millet Meclisi, 1932; Yildirim, 2007). Their proportion relative to the local population is illustrated in Table 2.3. It is important to note that this table captures only those migrants who arrived in the *exchangee* wave 1921-1927 between Greece and Turkey, as recorded in the source; it therefore excludes earlier muhacir waves and should be treated as a lower bound on the total migrant population settled in these provinces by the late 1920s.

Table 2.3 Migrant Population, Total Provincial Population, and Migrant Share by Province, 1927

Province	**Migrant Population**	**Total Provincial Population**	**Ratio (%)**
Balıkesir	37,088	421,066	8.81
Bursa	34,148	402,595	8.50
Çanakkale	10,856	181,735	5.97*
Edirne	49,336	150,840	32.71
İzmir	30,095	526,955	5.72
İstanbul	35,487	794,444	4.47
Kırklareli	27,254	108,989	25.01
Kocaeli	20,470	586,600**	7.14
Manisa	13,829	374,013	3.70
Niğde	15,671	166,056	9.44
Samsun	22,579	274,065	8.24
Tekirdağ	30,243	131,446	23.01

Notes: General Population Census of 28 October 1927 (28 Teşrin-i Evvel 1927 Umumi Nüfus Tahriri[a]): Mufassal Neticeler, İcmal Tabloları; McCarthy (1983). (1) Migrant population figures for 1927 are drawn from the Turkish Statistical Yearbook (Vol. III, 1930); provincial population figures are taken from the Republic of Turkey's first census (1927). (2) Migrant figures capture only those who migrated between Turkey and Greece under the Population Exchange between 1921 and 1927 (see Table 3.2); this figure should therefore be treated as a lower bound on the true number of migrants settled in these provinces. * The source table lists this value as 85.97%; recomputing from the reported migrant and total population figures yields 5.97%. ** Recomputing the reported ratio (7.14%) against the reported migrant population (20,470) implies a total provincial population of approximately 286,600, not 586,600 as printed.

a. *Teşrin-i Evvel* was the Rumi calendar name for October, used in official Ottoman and early Republican administrative dating.

Overall, historical evidence suggests that settlement decisions were shaped primarily by the administrative capacity of the Turkish state, logistical constraints, and reconstruction priorities rather than agricultural fit. In practice, the ministry's

stated objectives were repeatedly constrained by time pressures and the need for rapid execution of the plan, reconstruction priorities, and logistical considerations that drove settlement geography. As a result, a substantial share of *Lausanne exchangees* were settled in locations that did not necessarily correspond to their prior agricultural background. This generated meaningful variation in the degree of agricultural compatibility across settlements, which this thesis exploits as part of its quasi-experimental research design. Left-over vacant land, mismanaged property restitution, and informal seizure by residents created further economic precarity for the incoming (*Lausanne exchangees*). This variation in settlement compatibility, together with the economic precarity documented above, motivates the theoretical framework developed in the following chapter. This thesis builds its hypotheses on these historical dynamics and their relationship with subsequent agricultural and long-run political outcomes.

3. THEORETICAL FRAMEWORK

3.1 Literature Review

3.1.1 Short-Term Effects Of Forced Migration And Economic Disruption

Previous studies, focuses on the contemporary effects of forced migration, short-term outcomes, and the host society's response. The supply shock to the labor market posed by Syrian refugees has been studied in depth (Ceritoglu et al., 2017; Del Carpio & Wagner, 2015). The large inflow of Syrian refugees reduced informal employment for natives (Ceritoglu et al., 2017); this eventually led to large-scale displacement of natives and a demand shock for formal-sector jobs among natives (Del Carpio & Wagner, 2015). In parallel with these findings, Aksu et al. (2022) find that while the overall impact on native men's total employment is null, the refugee influx has strong adverse effects on natives in the informal sector, particularly temporary, less-educated, young, and agricultural or construction workers, while native workers in the formal sector, especially in manufacturing, see favorable effects on employment and wages: for every 10 Syrian refugees who arrive, approximately 6 formal-sector jobs are generated for native men. However, they also emphasize that these adverse effects were concentrated among women and low-educated workers, for whom there were few formal-sector jobs available (Aksu et al., 2022; Del Carpio & Wagner, 2015).

In contrast, from the refugee perspective, Syrian refugees integrated into the labor market with moderate success over time. According to Demirci & Kırdar (2023), approximately four years after arrival, 61.8% of Syrian men aged 18-59 are employed, compared to 68.9% of native men in the same age group. This labor market pressure also affects native households: Çakır, Erbay & Kırdar (2023) find that native

children's employment falls following refugee arrival, primarily as children shift from combining work and school toward school only, though this shift toward schooling is concentrated among boys, particularly those with more educated parents, while girls with less-educated parents show an increased likelihood of being neither employed nor in school.

However, all these effects on education and the labor market are not uniform across Turkey: the labor market supply shock in the informal sector is stronger in relatively less-developed regions, where refugees act as closer substitutes for native workers (Aksu et al., 2022). Overall, the literature on forced migration, economic disruption, and their economic and political consequences is rich, particularly regarding the impact on native populations.

On the other hand, economic disruption generates persistent political consequences even in the absence of forced migration. Autor et al. (2020) find that rising import competition from China contributed to the election of more ideologically extreme representatives in trade-exposed US congressional districts, with effects concentrated in majority-white, non-Hispanic constituencies. Similarly, import competition with China led exposed communities to vote for radical right parties across 15 European countries on a broader, cross-national scale (Colantone & Stanig, 2018). Nevertheless, this effect is not only observed following particular trade shocks but overall from economic disruption. Fetzer (2019) shows that UK austerity-induced welfare cuts significantly increased support for the UK Independence Party (UKIP), with areas most exposed to welfare reductions demonstrating the sharpest rise in anti-establishment voting. Overall, not only trade shocks but economic disruption and economic precarity of certain social classes affect their political preferences, especially their voting preferences.

Taken together, this literature demonstrates that forced migration and comparable economic disruptions generate immediate, measurable effects on both migrants and host populations. However, because these episodes, the Syrian displacement into Turkey, the 2022 Ukrainian migration into Poland, and the trade- and austerity-driven disruptions studied by Autor et al. (2020), Colantone and Stanig (2018), and Fetzer (2019), are either still unfolding or too recent to observe multi-generational outcomes, this body of work cannot yet answer whether the effects it documents are transient or durable. Isolating a genuinely long-run legacy requires either waiting several more decades to observe these same contemporary cases, or turning to a historical episode with comparable structural features but sufficient elapsed time.

3.1.2 Long-Term Effects Of Forced Migration And Economic Disruption

Previous literature has identified that forced migration reshapes the economic orientation of the refugees. Becker et al. (2020) document what they term the "uprootedness hypothesis": having lost all physical possessions in the post-WWII expulsion of Poles from the Kresy territories, forced migrants shifted their investment preferences away from vulnerable physical assets and toward human capital. Descendants of displaced Poles today have significantly higher educational attainment than other Poles, averaging roughly one additional year of schooling. Forced migration can also alter the economic habits of the displaced by breaking their attachment to a particular occupation or place. Sarvimäki et al. (2022) argue that many people remain in low-income agricultural sectors due to an emotional attachment to their birthplace, which they term "location capital". However, forced displacement breaks this bond, making individuals more responsive to higher-earning opportunities elsewhere: displaced Finnish farmers were significantly more likely to switch to manufacturing and move to urban areas, resulting in incomes 16-30% higher than those of non-displaced farmers (Sarvimäki et al., 2022). In other cases, refugees bring professional expertise that directly boosts economic outcomes in their destination. Murard & Sakalli (2020) report that incoming Orthodox refugees to Greece after the 1923 population exchange brought industrial knowledge, particularly in the textile and carpet industries, along with agricultural expertise in crops such as tobacco and cotton, resulting in higher earnings and industrialization sixty years later. Notably, prosperity peaked in municipalities with a roughly even mix of refugees and natives, suggesting that social interaction and knowledge-sharing mattered more than the sheer scale of the inflow. The 1947 Partition of India followed a related but distinct logic: migratory flows exhibited a "relative replacement effect," with in-migrants moving primarily into districts that had experienced the largest outflows of the minority group, alongside a strong pull toward areas near the border and large cities (Bharadwaj et al., 2008). The Turkish case appears to have followed a different logic: historical accounts suggest that settlement was driven primarily by logistical convenience and reconstruction priorities, and even the intended placement of *exchangees* in Orthodox-vacated properties was undermined by informal looting and property destruction, as mentioned in Chapter 2. Becker & Ferrara's (2019) survey of this literature identifies two broadly opposing patterns: mass inflows of expellees often accelerate structural transformation away from low-productivity agriculture toward manufacturing, generating agglomeration economies in receiving regions, while conversely, the mass expulsion of high-skilled minorities, such as Jews from the Soviet Union or managers from Nazi Germany, can produce

permanent reductions in a region's middle class, firm value, and long-run income. This pattern is also anecdotally observed in Turkey after the population exchange: the wine industry in Turkey suffered significant setbacks in its aftermath (Balta, 2017).

In addition, scholars also identified long-run political outcomes to be affected by historical disruptions. Voigtländer & Voth (2012) found that German towns that witnessed pogroms in the 14th century were more likely to vote for the NSDAP and engage in anti-Semitic violence. For the Lausanne case, Murard (2023) found that there was no political fragmentation or political instability in resettlement places. Politically, then, refugees appear well integrated in the long run. However, evidence from Grigoriadis & Moschos (2023) suggests that while this integration did not produce political instability, it was not ideologically neutral: refugee settlement produced a persistent, significant increase in support for left-wing parties overall, though this effect was considerably stronger for refugees originating from Asia Minor than for those from East Thrace and the Black Sea (Pontus) region. In the post-dictatorship period, these differences dissipate. Related literature demonstrates that settlement circumstances more broadly can generate durable political preferences, even without the specific mechanisms documented above. Bazzi et al. (2020) find that longer historical exposure to frontier settlement in the United States predicts stronger present-day opposition to redistribution and higher support for the Republican Party, an effect they attribute to both the selective self-sorting of individualist settlers into frontier areas and the direct effects of harsh, isolated settlement conditions. Because frontier settlement was voluntary, this mechanism is more difficult to separate from self-selection than in the compulsory Lausanne exchange, although settlement locations remained administratively determined.

A distinct but related literature addresses the mechanism by which economic conditions translate into voting behavior in the first place. Retrospective voting theory holds that voters evaluate incumbents primarily on the basis of conditions experienced during their tenure, rewarding or punishing the governing party accordingly, largely independent of its ideological positioning (Fiorina, 1981; Healy & Malhotra, 2013; Key et al., 1966). This theory, however, is typically formulated at the level of an individual voter's own recent experience, and does not on its own explain why a grievance originating in a single historical episode might continue to shape voting behavior across multiple subsequent generations. The historical persistence literature addresses this second question directly: shocks originating in the distant past can leave lasting imprints on political behavior through channels of cultural and intergenerational transmission, even after the material conditions that

originally produced them have faded. Voigtländer and Voth's (2012) finding that medieval anti-Semitic violence predicted twentieth-century Nazi voting is itself an instance of this broader pattern, as is Nunn's (2020) synthesis of how historical shocks persist through inherited institutions, norms, and community narratives long after their proximate economic causes have dissipated. Taken together, these two literatures suggest a plausible pathway by which a disruption experienced by Lausanne exchangees in the 1920s could continue to shape the political orientation of their descendants a century later, transmitted through community identity and historical memory rather than through directly inherited material grievance.

This long-run literature establishes that displacement-driven economic and political legacies can persist for many decades, and in some cases a full century, after the initial shock. It also reveals a further gap this thesis is positioned to address: few of these studies examine an economic outcome and a political outcome for the same displaced population using the same geographic unit of analysis, and fewer still test explicitly whether the former statistically mediates the latter. The 1923 Lausanne population exchange offers a case where this joint design is feasible, given the century of elapsed time and the availability of both historical agricultural statistics and modern electoral data at fine spatial resolution.

3.1.3 The 1923 Lausanne Population Exchange: Existing Evidence And The Gap

The 1923 Lausanne exchange sits precisely at the intersection of the short-term and long-term literatures reviewed above: it was, at the time, a forced migration crisis materially similar to the contemporary Syrian and Ukrainian cases discussed in Section 3.1.1, but with over a century having since elapsed, it now permits the kind of long-run, joint economic-political analysis called for in Section 3.1.2. A substantial body of scholarship examines the 1923 population exchange between Greece and Turkey. This literature is largely situated within diplomatic, historical, and sociological traditions, emphasizing the legal foundations of the exchange, its role in nation-building, and its social consequences for displaced populations. A separate and largely disconnected literature in economics studies the effects of forced displacement on host communities more generally, a field that traces to Card (1990) and the subsequent debate over identification strategy (Borjas, 2017), and that has expanded substantially since the 2011 Syrian refugee crisis (Verme & Schuettler, 2021). Recent syntheses of this literature find that forced displacement is, on av-

erage, neutral to positive for host community economic outcomes, with effects that tend to strengthen over the medium and long run as population shocks persist (Rozo & Grossman, 2025). This thesis situates the 1923 exchange within this broader economics literature, treating it as a historically plausible quasi-experimental case with the benefit of a century of downstream data, an advantage rarely available in the study of contemporary displacement episodes.

One strand of the exchange-specific literature focuses on the diplomatic and legal dimensions of the Lausanne Convention. These studies mostly examine the diplomatic setting in which the Convention was signed, how it was crafted, and how the decision-making process was conducted from a legal perspective (Alpan, 2024; Özsu, 2011; Yildirim, 2007). They provide comprehensive accounts of both the Greek and Turkish states' understanding of self-determination and their definitions of minority and nation, highlighting the importance of post-World War I international diplomacy and early twentieth-century bureaucratic conceptions of nationhood and nation-building in shaping the population exchange (Kulczycki, 2022). While this work provides valuable insight into the institutional and diplomatic origins of the exchange, it does not examine its long-run political consequences at the subnational level.

A second strand consists of historical and sociological accounts of the population exchange, which document its political, economic, and cultural effects on migrant communities (Hirschon, 2003; Lamprou, 2023; Yildirim, 2006; Yıldırım, 2023b). These studies offer rich qualitative evidence on the lived experiences of refugees and the short- and long-term challenges of resettlement, often from both international and refugee-centered perspectives. Hirschon (2003) notes, in particular, that the consequences of the exchange have received surprisingly little scholarly attention despite their considerable relevance to the present day. However, this literature remains primarily qualitative and does not systematically evaluate whether forced settlement generated persistent differences in economic or electoral outcomes.

Moreover, existing scholarship on the population exchange is asymmetrical, both in its empirical focus and in its methodology. On the Greek side, a growing quantitative literature examines the long-run economic and political consequences of refugee resettlement using precinct- and municipality-level data. Murard and Sakalli (2020) finds that Orthodox refugees resettled in Greece achieved successful economic integration within a few generations, while Murard separately finds that their political and civic integration was similarly successful, with no lasting negative effect on social cohesion. Kritikos (2005) and Kontogiorgi (2006) document that refugee resettlement in Greek Macedonia was accompanied by the transfer of agricultural land and

productive assets abandoned by departing Muslim populations, with almost none of this land redistributed to native Greek peasants, and that this transfer was administered with the assistance of the League of Nations' Refugee Settlement Commission, resulting in a steady and rapid expansion of cultivated land, a pattern of capital transfer directly analogous to the one this thesis investigates on the Turkish side. Michalopoulos (2017) traces the political identity of refugee neighborhoods in Greek urban centers over a full century, finding that a leftward political orientation among refugee-origin voters persisted through the Second World War and the restoration of democracy in 1974-75. This is consistent with Grigoriadis and Moschos (2023), discussed above, who similarly document a persistent positive effect of refugee settlement intensity on left-wing vote share at the sub-prefectural level. Studies centered on the Turkish experience remain comparatively limited (Lamprou, 2023), particularly with respect to quantitative analysis of political and economic outcomes at fine spatial resolution.

In contrast, a separate body of research employs quantitative methods to study minority groups in Turkey. These studies examine the political and economic effects of ethnic and religious minorities, most notably Greek and Armenian communities, using large datasets and fine-grained geographic controls (C. E. Arbatli & Gomtsyan, 2019; E. Arbatli & Gokmen, 2016, 2018). One closely related contribution examines outcomes at the district level in areas with a strong prior Greek or Armenian presence (E. Arbatli & Gokmen, 2018); this thesis differs in its finer neighborhood-level spatial resolution and its joint treatment of agricultural and long-run electoral outcomes. Nevertheless, this literature focuses on historically endogenous minority populations rather than groups formed through a discrete, state-mandated episode of forced migration.

Taken together, existing research offers rich qualitative accounts of the population exchange, rigorous quantitative analyses of minority politics in Turkey, and a maturing quantitative literature on the Greek side of the exchange, but these literatures remain largely disconnected. To the best of my knowledge, no existing study combines fine-grained spatial data, an agricultural production outcome, and a century-long electoral outcome for the Turkish side specifically. There remains limited systematic, quantitative evidence on whether the forced settlement of muhacirs following the 1923 population exchange generated persistent differences in electoral behavior, and the extent to which agricultural production may constitute one potential pathway. This thesis examines this gap by integrating historical settlement data with subnational agricultural and electoral outcomes to assess the economic and long-run political legacy of forced migration in Turkey.

3.2 Theoretical Framework

The compulsory exchange created a large, externally imposed displacement shock, while the geographic allocation of settlers remained administratively determined and potentially selective. The 1923 Lausanne Convention constitutes a case of a historically plausible quasi-experimental demographic shock with potential long-term political consequences that can be observed. Political economy theories and studies suggest such exogenous shocks can reshape economic structures, material interests, human capital, and political behaviour. However, when displaced populations are resettled under uniform political institutions and experience integration and domestic migration, early effects may attenuate. The 1923 population exchange presents a theoretically ambiguous case, worth empirical investigation.

Muslim refugee (*Lausanne exchangees*) settlement was a state-coordinated settlement process, limiting individual preference over location. Initial settlement locations were determined by the state officials with regard to reconstruction, repopulation, and agricultural purposes (Arı, 1995). Nevertheless, in practice, the implementation never fully followed the state instructions and produced heterogeneity in settlement decisions. This heterogeneity suggests that arriving *Lausanne exchangees* could plausibly have experienced one of at least three distinct material trajectories:

(i) *economic disruption*, arising from a mismatch between settlers' prior agricultural expertise and local conditions;

(ii) *economic benefit*, arising from inherited productive capital left behind by the departing population; or

(iii) *destruction of inherited capital*, arising from the degradation of inherited property through neglect or looting by residents before settlers' arrival.

Because the observable consequence of trajectory (iii) is a reduction in agricultural production similar to that implied by trajectory (i), the two are treated jointly in the discussion of the disruption channel below, rather than as separately identifiable pathways.

Incomplete restitution of property likely generated an initial negative wealth shock for many *Lausanne exchangees*. This asset loss may alter the household's economic security and material interests. Moreover, the mismatch of settlement locations' agricultural suitability with migrants' prior know-how presented additional

challenges, potentially affecting agricultural production after migration. Reduced production and adjustment costs may have affected local economic trajectories in *Lausanne exchangees* settled areas. Additionally, there were reported seizures of Orthodox property before the arrival of refugees (Arı, 1995; Köker & Keskiner, 2003). This process of informal expropriation was compounded by broader housing shortages in Anatolia, generating substantial disruption to property relations more generally (Aktar, 2003). Apart from the property seizures of locals, in some cases incoming refugees (*Lausanne exchangees*) may also have contributed to this process by destroying the inherited agricultural infrastructure due to the limited knowledge of crop expertise (Köker & Keskiner, 2003). This destruction of inherited capital, however, is more persistent in tree-crops (e.g., olive/grape) rather than annual crops (e.g., tobacco), hence production gaps in tree-crops would reflect genuine capital destruction rather than a mismatch between settlers' prior expertise and local conditions.

The economic disruption channel, while central to this framework, is not the only plausible material mechanism through which the exchange may have shaped local economic trajectories. An alternative and competing channel operates through the transfer, rather than the loss, of productive capital. Because the exchange coincided with the departure of Orthodox Christian communities who held distinct agricultural specializations, most notably in olive cultivation and viticulture, arriving refugees (*Lausanne exchangees*) in many localities inherited standing agricultural assets: mature groves, established vineyards, and existing structures that could not have been rapidly replicated regardless of the settlers' own prior expertise. Under this channel, local economic outcomes might depend less on the match between settlers' skills and local conditions than on the extent of productive capital left behind by departing populations, a mechanism documented in the Greek context by Kritikos (2005) and Kontogiorgi (2006). A further complication specific to viticulture is that the departure of the primary wine-producing population coincided with domestic prohibition policy in Turkey (the 1920 Men-i Müskirat Kanunu), such that any inherited viticultural capital may have persisted in the form of continued grape cultivation for non-alcoholic uses even as wine production itself did not survive the exchange, a distinct institutional channel operating independently of settler skill or capital transfer. However, the scholarly literature on capital transfer during the Lausanne Population Exchange is largely Greek-side specific (Kontogiorgi, 2006; Kritikos, 2005). Moreover, for this channel to operate, the inherited property and agricultural infrastructure needed to have remained intact, despite harm caused by departing populations and seizures by local residents (Köker & Keskiner, 2003). Therefore, this channel might not hold as reliably on the Turkish side of the Lau-

sanne Population Exchange. Because settlement outcomes plausibly varied across localities, the disruption and capital-inheritance channels described above are not mutually exclusive at the national level: different refugee (*Lausanne exchangees*) communities may have experienced different dominant channels, depending on local conditions. The neighborhood-level granularity of the data used in this thesis allows heterogeneity in estimated production effects to be examined, rather than averaged away in a single national estimate, although, as discussed above, distinguishing the specific underlying mechanism behind a given local outcome remains possible only in limited cases, such as the tree-crop durability test described in the previous section.

This economic precarity may force migrants to change their occupations and influence long-term political preferences indirectly. Under the disruption channel described above, these households may develop distinct preferences regarding certain policies due to economic disruption after migration. Under the capital-inheritance channel, by contrast, no comparable material grievance need arise, and any political divergence would instead have to operate through a separate identity- or community-based mechanism rather than through material hardship. Evaluating the relative importance of these channels and establishing whether any observed political legacy is in fact mediated by economic outcomes at all is a central empirical objective of this thesis.

A null or convergent result in observed economic outcomes 15 years after resettlement would be consistent with two distinct underlying stories: either no meaningful disruption ever occurred, or disruption did occur but was overcome over time through adaptation, learning, or the passage of time. These two stories carry very different implications for the political legacy. The former would suggest no material grievance ever existed to generate systematically different political outcomes; the latter would suggest a real grievance existed early on, whose economic trace has since dissipated even as its political trace has not. This ambiguity motivates the use of formal mediation analysis as an exploratory assessment of empirically important issues: this thesis therefore tests whether economic outcomes mediate the relationship between settlement and long-run political outcomes, rather than relying on economic outcomes alone to establish whether a material channel was ever active.

However, these preferences may disappear or attenuate due to high levels of integration and domestic circumstances. Urbanization, social integration, domestic migration, and nation-building efforts may disrupt the channels that allow these preferences to be embedded in local refugee (*Lausanne exchangees*) communities. Whether these political differences persist is an empirical question rather than a theoretical certainty.

This framework applies to cases of state-mandated compulsory migration involving limited administrative capacity, imperfect compensation, and heterogeneous settlement outcomes. Taken together, this framework yields three testable hypotheses.

H1 (Production Effect). If settlement was primarily shaped by administrative logistics, proximity to rail infrastructure and the availability of vacated Orthodox housing — rather than a match between settlers' prior agricultural expertise and local conditions, agricultural production in localities exposed to *Lausanne exchangees* settlements is expected to be *lower* than production in comparable non-*exchangee* localities.

This hypothesis is tested across four comparisons, each isolating a distinct component of the exchange: **Two-way Exchange vs. Control**, **One-way Orthodox Departure vs. Control**, **One-way Muslim Arrival vs. Control**, and **Two-way Exchange vs. One-way Orthodox Departure** (isolating the marginal contribution of settler arrival). A negative effect concentrated in Two-way Exchange and One-way Muslim Arrival would be consistent with settler skill mismatch, specifically as the operative channel. A negative effect concentrated in One-way Orthodox Departure would instead implicate the loss of Orthodox agricultural expertise and infrastructure, independent of settler characteristics. A more compehensive explanation for these comparison groups is provided in Section 4.1.

Should the estimated effects instead be positive and significant, this would instead be more consistent with the rejection of H1 in favor of an alternative mechanism, such as the transfer of standing agricultural capital (mature olive groves, established vineyards, and cultivated land) from departing Orthodox communities to arriving settlers.

H2 (Political Opposition Effect). Neighborhoods historically exposed to the settlement of *Lausanne exchangees* are expected to exhibit greater electoral support for opposition political parties relative to comparable non-*exchangee* exposed localities.

Building on retrospective voting theory (Fiorina, 1981; Key et al., 1966) and the historical persistence literature on the intergenerational transmission of political attitudes (Nunn, 2020; Voigtländer & Voth, 2012), this framework predicts a specific direction: because the material and social disruptions associated with forced resettlement were experienced under the administrative authority of the state responsible for the exchange, and because retrospective voting theory holds that voters attribute responsibility for adverse conditions to governing authority, this disruption

is expected to have generated a persistent political grievance directed at incumbency itself, rather than at any particular party or ideological position. If this grievance was transmitted across generations through the channels of cultural and historical memory documented in the historical persistence literature, neighborhoods historically exposed to Lausanne exchangee settlement are expected to exhibit systematically lower support for incumbent governing parties, and correspondingly higher support for opposition parties, relative to comparable non-exchangee exposed localities, in each electoral period examined, regardless of which specific party held government at that time.

Because the identity of the incumbent party changes across the two electoral periods examined in this thesis (Motherland Party [ANAP] in 1991; Justice and Development Party [AKP] in 2023), while the secular-versus-religious-conservative axis of Turkish party competition has remained comparatively stable across this period. The primary empirical test evaluates whether historically exposed localities exhibit higher opposition vote shares than otherwise comparable non-*exchangee* exposed localities. Party-specific analyses are subsequently presented to identify which political parties contribute to the aggregate opposition effect and to assess whether this persistent opposition orientation is accompanied by a broader ideological realignment.

H3 (Economic Mediation). If the persistent opposition orientation predicted under H2 arises, at least in part, from the economic disruption described under H1, agricultural production should statistically account for a positive portion of the relationship between historical exposure to *Lausanne exchangees* settlement and opposition vote share. Conversely, if the political legacy operates primarily through community identity, historical memory, institutional persistence, or other non-material mechanisms, the estimated indirect effect through agricultural production should be limited relative to the total political effect.

The statistical hypothesis concerns whether the estimated indirect effect through agricultural production is positive. The substantive interpretation, however, also depends on the magnitude of the estimated ACME relative to the total effect identified under H2. A positive but comparatively small mediated share would suggest that agricultural production statistically accounts for only a limited portion of the observed opposition effect. Such a result would be consistent with the importance of additional mechanisms not captured by the mediation model, although it would not by itself identify those mechanisms.

Because the mediation analysis relies on strong assumptions regarding treatment-mediator and mediator-outcome confounding, it is interpreted as an assessment of one candidate pathway rather than as a definitive identification of the underlying causal mechanism. These hypotheses are not mutually exclusive. Together, they evaluate the total political relationship, one candidate economic pathway, and the extent to which that pathway statistically accounts for the observed political association, rather than presupposing a single underlying mechanism. H1 tests whether *Lausanne exchangees* settlement affected agricultural production, expecting a negative correlation. H1 evaluates whether historical exposure to *Lausanne exchangees* settlement is associated with lower agricultural production, consistent with the economic-disruption argument. H2 evaluates whether historically exposed localities exhibit greater support for opposition political parties relative to otherwise comparable non-*exchangee* localities. H3 evaluates whether differences in agricultural production statistically account for part of the relationship identified under H2. Overall, these hypotheses distinguish between the existence of a long-run political legacy and one potential economic pathway through which that legacy may persist.

4. DATA

This study assembles a novel dataset linking neighborhood-level settlement records, agricultural production statistics, and geographic controls to examine the long-run relationship between historical settlement exposure during the 1923 Lausanne population exchange and agricultural production. The analysis operates at the neighborhood (*mahalle*/*köy*) level, the finest administrative unit available in Turkish records, covering 51,297 geographic units across Turkey.

4.1 Settlement And Treatment Data

The primary source for identifying exchange-affected neighborhoods is the Nişanyan Yeradları settlement database (Nişanyan, 2025) [1]

According to the community-label definitions provided on the Nişanyan (2025) website, I classify settlements into three distinct groups, which later form the basis of the treatment variable. Settlement-level community composition is recorded in the database as a set of categorical ethnic and religious group labels per settlement, without an explicit temporal dimension. I construct treatment status using theory-driven keyword classifications of these labels into three groups:

1. Nişanyan Yeradları: Inventory of Settlements in Turkey and Surrounding Countries, an online database of over 75,500 mapped place-names covering Turkey and neighboring countries (accessed April 2025). The dataset was made available for use in this thesis by Assoc. Prof. Mert Moral. A comprehensive catalogue of Turkish settlements containing information on the ethnic and religious community composition of each settlement historically. The database records up to four community group labels per settlement in the early 20th century, covering 2,151 settlements containing labels associated with the population exchange.

- **Exchangee core**: settlements where (Greece Immigrant) is present, which the database's own categorization restricts to arrivals under the 1923-24 Lausanne exchange specifically.
- **Exchangee broad (Balkan refugee)**: settlements where any Balkan Muslim ethnicity is present, migration beyond the subset of Lausanne-specific exchange. Consistent with this thesis's focus on Balkan-origin (*Rumeli*) muhacirs established in Chapter 2, this classification excludes labels associated with Tatar and Caucasian-origin migration, which reflect a historically and geographically distinct migration wave.
- **Previously Orthodox**: settlements where any Orthodox ethnicity departed. This classification assumes that the presence of a historical Orthodox label reflects genuine community presence before departure rather than incomplete database coverage; while this assumption cannot be directly verified against an independent source, it is broadly consistent with the documented scale of Orthodox displacement discussed in Chapter 2 (Behar, 1996).

I treat the presence of a *exchangee core* label together with a *previously Orthodox* label at the same settlement as the primary indicator of direct exchange (group Two-way Exchange); absence of a corresponding population currently recorded under Orthodox labels is interpreted as being consistent with the historical departure of those communities under exchange.

I define five mutually exclusive treatment groups at the settlement level:

- **Two-way Exchange:** *Exchangee core* present and at least one *Previously Orthodox* label present. Core Lausanne exchange group. ($n = 863$ neighborhoods)
- **One-way Muslim exchangee arrival:** *Exchangee core* present but no *Previously Orthodox* labels present. ($n = 49$ neighborhoods)
- **Broader Balkan Migration:** *Exchangee broad* labels present, with or without *Previously Orthodox* labels. Excluding Exchangee Core groups. Used in robustness analysis only.
- **One-way Orthodox Departure:** At least one *Previously Orthodox* label present but no *Exchangee core* label present. Designed to approximate the departure effect in isolation. ($n = 416$ neighborhoods)

- **No Exchange Record:** Not present in the exchange database. Primary control pool. ($n = 49{,}546$ neighborhoods)

Settlement-level classifications are aggregated to the neighborhood level. Neighborhoods containing settlements with conflicting group classifications are flagged as *mixed neighborhoods, n=82* and excluded from the primary analysis, though included in robustness checks.

4.2 Geographic Units

The geographic foundation of the analysis is the Turkish Address Registry, a comprehensive polygon dataset covering all 51,297 *mahalle* and *köy* units in Turkey. Geometries were precision-snapped to a resolution of 10^{-6} degrees and validated to correct topological errors. This process resulted in the exclusion of a single geographic unit from the original dataset.

To link modern neighborhood boundaries to historical 1935 administrative districts, I dissolve neighborhoods to the modern district level and join to the Livny (2024) crosswalk, which maps modern Turkish districts to their historical 1935 counterparts. This bottom-up approach is required because modern neighborhood boundaries have no direct historical predecessor; the crosswalk instead operates at the district level, where consistent historical correspondences exist. This procedure reconstructs 420 historical 1935 districts reconstructed from modern neighborhood geometries, with complete (100%) coverage across all matched districts as Figure 4.1 shows. Hatay is included in the map despite not being part of Turkey's territory in 1935, as it was annexed only in 1939 following the dissolution of the French Mandate's Sanjak of Alexandretta. It is retained here because the neighborhood assignments indicate the presence of *exchangee broad* settlement associated with the broader Balkan muhacir wave, and because post-1939 agricultural production statistics are available for it and are included in the analysis.

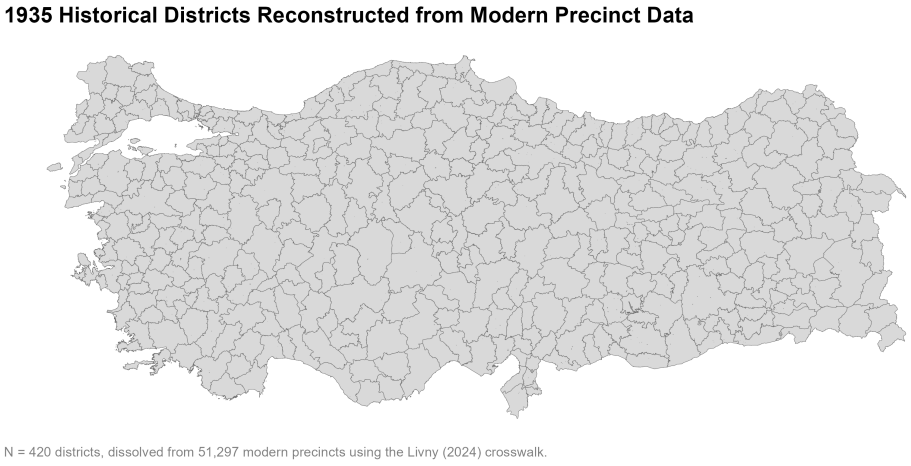


Figure 4.1 Reconstructed 1935 Historical Districts.

4.3 Agricultural Production Data

Agricultural production data are drawn from four separate official statistical publications of the late Ottoman and early Republican period, which report crop statistics at the province and at the district level, respectively. I digitize data for four crops selected to reflect the key agricultural specializations of both the departing Orthodox and arriving settler communities. Data for all four crops (olive, grape, tobacco, and wheat) are available for both the pre-treatment (1909, 1913, 1914) and post-treatment (1936-1946) periods, allowing consistent comparison of production outcomes across the treatment window for each crop. Throughout the empirical analysis, production variables are transformed as $\ln(\text{production}+1)$ to reduce skewness while retaining observations with zero production.

4.3.1 Pre-Treatment Agricultural Production Data

In 1909, 1913, and 1914, the Ottoman State kept agricultural records of the total production of various crops (Güran, 1997). I digitized and geocoded production

statistics of four crops at the province level for 1909, 1913, and 1914. Province-level production totals are matched to historical provinces via the Güran (1997) crosswalk provided within the document after normalizing province and district names. Historical province names are recoded to modern equivalents where necessary. The statistics do not include Kars, Ardahan, and Artvin. These provinces were under Russian administration until the end of World War I. Using the province (*vilayet*) and district (*kaza*) correspondence rules provided in the source, 83.3% of modern districts could be matched to their historical counterparts. The provinces that are mapped can be seen in Figure 4.2.

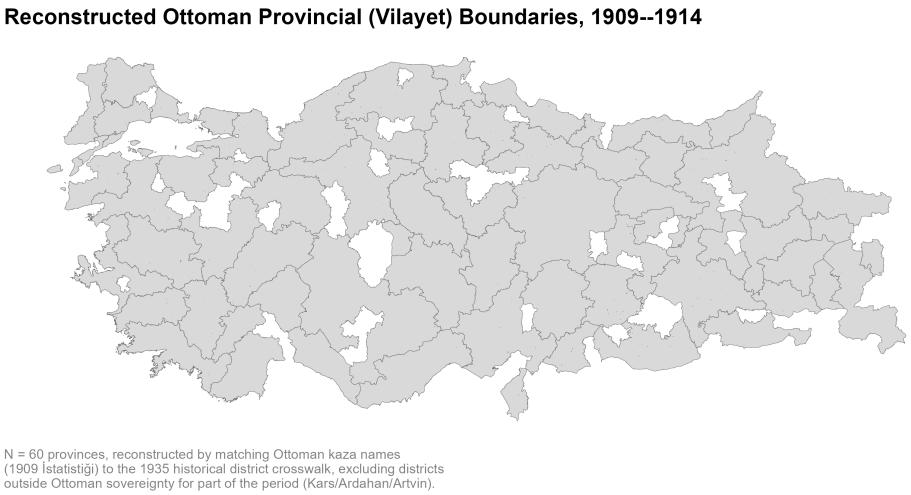


Figure 4.2 Reconstructed 1909 Historical Provinces.

- **Olive** (1909, 1913, 1914): Number of trees, cultivated area in hectares, and total production in tons.
- **Grape** (1909, 1913, 1914): Cultivated area in hectares and total production in tons.
- **Tobacco** (1909, 1913, 1914): Cultivated area in hectares and total production in tons.
- **Wheat** (1909, 1913, 1914): Cultivated area in hectares and total production in tons.

4.3.2 Post-Treatment Agricultural Production Data

In addition, I have digitized and geocoded early Republican statistical yearbooks on the same four crops as post-treatment observations.

- **Olive** (1936-1944, 10 annual observations): Total production in tons, number of trees, food-use olive, olive oil, extracted oil, and total value. Covers 120 districts. Olive cultivation constituted one of the principal agricultural activities historically associated with Orthodox communities in Western Anatolia, with trees requiring 15-20 years to reach full production (T.C. Başvekalet İstatistik Umum Müdürlüğü, 1943).
- **Grape** (1938-1942, one 5-year average): Total harvest in tons. Viticulture was another major agricultural activity historically associated with Orthodox communities, particularly in the Aegean and Thrace regions. Because wine production played a substantially larger role in Orthodox agricultural economies than among the incoming Muslim settlers, grape production provides a useful setting for examining the relative importance of cultural practices and inherited agricultural infrastructure (Başbakanlık İstatistik Genel Müdürlüğü, 1948).
- **Tobacco** (1944-1946, 3 annual observations): Total production in tons and cultivated area in hectares. Tobacco was a primary crop of Balkan Muslim communities from Greece and Bulgaria, making it a useful test case for settler human capital transfer (T.C. Başbakanlık İstatistik Genel Müdürlüğü, 1947).
- **Wheat** (1944-1946, 3 annual observations): Total production in tons and cultivated area in hectares. Wheat was used as a robustness check since it is not a specialty crop for both incoming or outgoing groups (T.C. Başbakanlık İstatistik Genel Müdürlüğü, 1947).

District-level production totals are matched to historical 1935 districts via the Livny (2024) crosswalk after normalizing province and district names.

4.4 Distributing District And Province Production To Neighborhoods

Since production data are available only at the province and the district level, I distribute province and district totals to constituent neighborhoods using a within-province/district share based on the estimated 1909/1935 agricultural land coverage. The allocation procedure, therefore, assumes that, within each historical district or province, agricultural production was proportional to each neighborhood's share of total agricultural land. This assumption is not directly testable with the available historical data and should therefore be interpreted as a measurement assumption rather than an empirical finding.

To estimate 1909 and 1935 neighborhood-level agricultural land, I use the CORINE Land Cover dataset, which provides harmonized land cover classifications at 100-meter resolution for five time points: 1990, 2000, 2006, 2012, and 2018 (Copernicus Land Monitoring Service, 2020a, 2020b, 2020c, 2020d, 2020e). For each neighborhood, I extract the total area classified under 11 agricultural land cover categories (CORINE codes 211–244) across all five years and fit a linear time trend. I then extrapolate this trend to 1909 and 1935 to obtain an approximation of the neighborhood-level estimate of agricultural land ($\widehat{\text{ag}}_{i,year}$).

Within each historical district or province d and baseline *year* (1909 or 1935), the neighborhood share of total agricultural land is:

$$s_i = \frac{\widehat{\text{ag}}_{i,year}}{\sum_{j \in d} \widehat{\text{ag}}_{j,year}} \tag{4.1}$$

Neighborhood-level production for neighborhood i in year t is then:

$$y_{it} = s_i \times Y_{dt} \tag{4.2}$$

where Y_{dt} is district or province d's total production in year t. I apply a uniform total agricultural land share across all four crops rather than crop-specific shares to avoid introducing crop-specific information that could mechanically reinforce the estimated treatment effects. Using post-exchange land cover to distribute pre-treatment production would risk mechanically reinforcing the relationship this

thesis seeks to test. Sanity checks confirm that $\sum_{i \in d} y_{it} = Y_{dt}$ for all district-year combinations, with ratios of exactly 1 throughout.

This approach relies on two measurement assumptions that are worth stating explicitly. First, extrapolating a linear trend fitted to CORINE observations from 1990 to 2018 (given the limited number of observed years, a high-order polynomial specification was not pursued) backward to 1909 and 1935 assumes that within-neighborhood agricultural land shares changed smoothly over this much longer period, which cannot be directly validated before 1990. Second, distributing production using a single agricultural land share, rather than crop-specific shares, implicitly assumes no systematic within-district or within-province variation in crop-specific comparative advantage across neighborhoods; production differences arising from such variation would not be captured by this method. Although this allocation inevitably introduces measurement error, there is little reason to expect such error to be systematically related to historical settlement status conditional on the matching covariates used in the empirical analysis. The linear trend itself is estimated separately for each neighborhood, rather than as a single national or regional trend, in order to preserve neighborhood-level heterogeneity in land-cover trajectories; Figure A.1 and Table A.2 illustrate this procedure using the national mean across neighborhoods for visual clarity, though the underlying estimation and back-projection are conducted at the individual neighborhood level. The linear specification, while a strong assumption over a 55-109 year extrapolation window, was retained as the most parsimonious specification that avoided this class of infeasible projections. Although this allocation inevitably introduces measurement error, there is little reason to expect such error to be systematically related to historical settlement status conditional on the matching covariates used in the empirical analysis.

4.5 Electoral Data

I have collected neighborhood-level election results for the 2023 presidential and parliamentary elections, and the 1991 parliamentary election. These represent, respectively, the most recent and earliest years for which neighborhood-level electoral results could be obtained. The 2023 results include both the first and runoff rounds of the presidential election as well as the general parliamentary election. The 1991 results include the general parliamentary election, also at the neighborhood level.

Following Grigoriadis and Moschos (2023), I construct party and coalition vote-share outcomes rather than a single composite ideology index, and following Murard (2023), I additionally construct turnout as an outcome variable. For 2023, outcomes are governing party AKP (Justice and Development Party) vote share, and main opposition party CHP (Republican People's Party) vote share, Cumhur İttifakı (People's Alliance) coalition vote share, Millet İttifakı (Nation Alliance) coalition vote share, and turnout, alongside first- and second-round vote shares for the two major presidential candidates, Kılıçdaroğlu and Erdoğan. The principal outcome for the 2023 parliamentary election is the vote share of the Millet İttifakı (Nation Alliance), which serves as the operational measure of electoral support for the opposition. Because Millet İttifakı combined CHP with ideologically heterogeneous partners, including the nationalist İYİP and the Islamist Saadet Partisi, its vote share is operationalized as the opposition to the incumbent coalition.

For 1991, party vote shares (SHP, DSP, DYP, ANAP, RP) and the turnout rate are the outcomes.

The 2023 neighborhood-level election results were matched to the geocoded neighborhood dataset at a 99.9% rate, and the 1991 neighborhood-level election results were matched at an 82.8% rate. The lower match rate for 1991 reflects province, district, and neighborhood name and boundary changes over the intervening three decades.

4.6 Control Variables

I construct a rich set of pre-determined control variables at the neighborhood level. All controls are either time-invariant geographic characteristics or measurements from the pre-treatment period that cannot have been affected by the 1923 exchange, except for nightlight luminosity and population density, discussed below. The full controls dataset covers all 51,297 neighborhoods with zero missing values.

4.6.1 Geographic And Location Controls

Neighborhood centroids are computed in EPSG:3035 metric projection. Neighborhood area is measured in km^2. Distance to the coastline is measured in kilometers using Natural Earth 10-meter coastline data. Seven geographic region dummies capture broad geographic, climatic, and developmental zone differences: Marmara, Aegean, Mediterranean, Black Sea, Central Anatolia, Eastern Anatolia, and Southeastern Anatolia.

4.6.2 Market Access

Distance to the nearest major port is computed as the shortest distance to the coordinates of 15 major Turkish ports listed in the 1980 Kabotaj İstatistikçi (Kabotage Statistics) yearbook (Başbakanlık Devlet statistik Enstitüsü, 1986), the earliest official government publication listing major Turkish ports that I was able to identify. Distance to pre-1923 railway lines is computed from a manually curated set of OpenStreetMap railway segments corresponding to lines constructed before 1923, cross-referenced against historical railway company records (British ORC, French SCP, German CFOA, and Baghdad Railway) (Trains of Turkey, 1918). Distances to Istanbul, Ankara, and İzmir are computed as distances from neighborhood centroids.

4.6.3 Terrain

Elevation, slope, terrain ruggedness index (TRI), and surface roughness are extracted from the SRTM 90-meter digital elevation model via the `elevatr` package (Hollister et al., 2025). Terrain derivatives are computed using the `terra` package with 8-neighbor windows. For 415 coastal neighborhoods (0.8% of total) whose centroids fall on sea pixels with no valid elevation data, I assign elevation = 0 and fill slope, TRI, and roughness using focal neighborhood means from adjacent land pixels, consistent with their confirmed coastal or sea-level locations.

4.6.4 Agricultural Suitability

Crop suitability indices for olive, tobacco, and wheat are extracted from the FAO GAEZ v4 dataset (FAO & IIASA, 2021) at approximately 1-kilometer resolution, using rainfed, low-input, 1961-1990 climate baseline scenarios. GAEZ suitability indices are used as a time-invariant measure of agro-ecological potential independent of actual crop choices, making them appropriate pre-treatment controls.

4.6.5 Pre-Treatment Climate

Mean annual temperature and total annual precipitation for the pre-treatment period 1901-1922 are derived from the CRU TS 3.22 dataset (University of East Anglia Climatic Research Unit et al., 2014) at 0.5-degree resolution (≈50 km). Monthly values are averaged over all 264 months in the 1901–1922 period. Given the coarse resolution of CRU TS relative to neighborhood size, these controls capture broad agro-climatic zone differences across Turkey rather than fine-grained local climate variation.

4.6.6 Distance To District Center

Distance to the 1935 historical district administrative center is measured in kilometers from each neighborhood centroid to a manually geocoded set of district center coordinates.

4.6.7 Nightlight Luminosity

I calculate the mean value of the VIIRS 2022 annual composite index (Elvidge et al., 2021) as a proxy for socio-economic development for each neighborhood. Nightlight luminosity is widely used as a proxy for local economic activity and socioeconomic development.

4.6.8 Population Density

I proxy population density at the neighborhood level by combining electoral data with geocoded neighborhood data, using the registered number of voters per neighborhood and the total neighborhood area to construct a proxy for urbanization.

Nightlight luminosity and population density are contemporary (2022-2023) measures and are therefore included only in the 2023 electoral specifications, where they serve as controls for present-day socioeconomic development. They are not used in the 1991 electoral analysis, nor in the agricultural production analysis, since they postdate those outcomes by decades.

Descriptive statistics for all control variables, by treatment group and before matching, are reported in Table **??**.

5. METHODOLOGY

The forcibly displaced people from Greece after the 1923 Population Exchange were resettled by a compulsory state-mandated procedure. Hence, observable pre-treatment characteristics can still differ systematically even in the absence of individual self-selection. To improve covariate comparability across treatment groups, I employ two complementary matching algorithms. Entropy balancing serves as the primary method, since it reweights rather than discards control units, preserving the full treated sample, an important property given the relatively small size of several treatment groups (e.g., n=49 for One-way Muslim Arrival). Unlike propensity-score matching, entropy balancing directly optimizes covariate balance over specified moments rather than relying on estimated treatment probabilities. Coarsened exact matching serves as a robustness check, since its exact-matching-within-bins logic guarantees covariate balance within the matched sample, at the expense of discarding unmatched observations. Entropy balancing is preferred as the primary specification because it retains the complete treated sample while achieving exact balance on specified covariate moments. CEM is reported as a robustness check because it imposes stricter common-support requirements at the cost of discarding unmatched observations. Finally, I estimate Difference-in-Differences specifications to account for time-invariant differences between treated and comparison units, conditional on the parallel trends assumption discussed below.

For electoral analysis, I used EB matching with comprehensive covariates, including contemporary measures for economic development and urbanization, to analyze the electoral behavior of the treated neighborhoods.

5.1 Identification Strategy

Although the population exchange was administratively compulsory, this does not imply that treatment assignment was as good as random. Settlement locations were shaped by logistical convenience, rail access, and the availability of vacated housing, all of which plausibly correlate with a neighborhood's pre-existing agricultural and geographic characteristics. A purely experimental framing is therefore unavailable, and this thesis does not have access to an instrument or a discontinuity that would support an instrumental-variables or regression-discontinuity design. The identification strategy pursued here instead rests on a selection-on-observables assumption: that, conditional on the geographic, climatic, and market-access covariates, treatment assignment is independent of potential outcomes.

This assumption motivates matching rather than a simple ordinary least squares regression on the full sample. Where treated and control neighborhoods are poorly balanced on observed covariates, OLS relies on the correctness of its linear functional form to extrapolate the counterfactual outcome in regions of covariate space with little or no common support. Matching methods instead construct an explicitly comparable estimation sample before estimating any treatment effect, reducing this extrapolation risk. Entropy balancing and coarsened exact matching are used together because they make different trade-offs along this dimension: entropy balancing retains the full treated sample by reweighting controls, at the cost of some residual reliance on the reweighting model's specification, while coarsened exact matching guarantees exact covariate balance within matched bins, at the cost of discarding a substantial share of the treated sample. Agreement between the two methods provides evidence that the estimated effects are not an artifact of either specific weighting or matching procedure.

The selection-on-observables assumption underlying both matching methods is only as credible as the covariate set used to implement it, and this thesis directly probes this assumption rather than asserting it. A placebo check applying the same entropy balancing procedure to pre-treatment (Ottoman-era) agricultural outcomes shows that the available covariates do not fully equate treated and control neighborhoods on outcomes observed *before* the exchange occurred. [1], indicating that cross-sectional matching alone cannot be treated as fully identifying the causal effect of the exchange. This finding directly motivates the inclusion of the difference-in-

1. See Appendix Table A.3

differences design: rather than requiring the stronger assumption that treatment is unconfounded conditional on observables, difference-in-differences requires only that any remaining confounding be time-invariant, which is a substantially weaker and more defensible assumption in this setting.

5.2 Treatment And Comparison Groups

The primary treatment is group Two-way Exchange: Greece Immigrant in, Greek Orthodox out. In addition, there are four comparison groups to isolate the effect of different conditions on the neighborhood's agricultural performance.

- **Two-way Exchange vs. Control:** main estimation of direct exchange relative to the control neighborhoods where there are no exchanges
- **One-way Orthodox Departure vs. Control:** comparison group to approximate the effect of Orthodox departure
- **One-way Muslim Arrival vs. Control:** comparison group to approximate the effect of incoming Muslim immigrants without prior Orthodox agricultural infrastructure
- **Two-way Exchange, vs. One-way Orthodox Departure** direct exchange compared to Orthodox infrastructure effect

My main analysis sample is restricted to clean neighborhoods and Lausanne Exchange refugees (*Lausanne exchangees*), where there is only one type of settlement. The Balkan Refugee groups' and mixed neighborhoods analysis are reported separately as robustness in the Appendix.[2]

Standard errors are clustered at the neighborhood level throughout this thesis, across all specifications: entropy balancing, coarsened exact matching, difference-in-differences, and formal mediation analysis. This choice reflects the finest level at which treatment status is assigned and observed: because all covariates, matching weights, and outcomes are constructed at the neighborhood level, neighborhood-level clustering is the most conservative choice available given the data structure,

2. For agricultural analysis see Tables A.12, A.13, A.14, A.15, A.16, A.17, and for electoral analysis see Tables A.36, A.37, A.43, A.44, A.38, A.39, A.51, A.52.

and does not require assuming independence of unobserved shocks across observations sharing a common historical district or province (Abadie et al., 2023). District- or province-level clustering was not adopted as the primary specification because the number of historical districts (420) and provinces (approximately 60) is small enough that cluster-robust variance estimation at these coarser levels becomes unreliable, particularly for treatment groups such as One-way Muslim Arrival (n=49), where the number of clusters containing treated units would be a small fraction of the total. Neighborhood-level clustering is therefore used uniformly, with the caveat that it may understate standard errors to the extent that unobserved shocks are correlated across neighborhoods within the same district or province rather than being neighborhood-specific.

5.3 Primary Method - Entropy Balancing

I employed Entropy Balancing as the matching method, which reweights the control observations to match the means of the treated group based on a pre-specified set of observed covariates. 14 different covariates are used:

- Elevation
- Slope
- Terrain Ruggedness index
- Distance to coast
- Neighborhood area
- Olive suitability index
- Tobacco suitability index
- Wheat suitability index
- Mean temperature between 1901 and 1922
- Mean precipitation between 1901 and 1922
- Distance to nearest port

- Distance to railways
- Distance to nearest major city
- Distance to historical district center

Entropy Balancing does not discard any treated observation while matching, hence more suitable to prevent loss of treated units. The post-matched covariates are perfectly balanced, as the standard mean error of all 14 covariates is 0 by construction. [3]

5.3.1 Robustness: Coarsened Exact Matching

As a robustness check, we replicate the analysis using Coarsened Exact Matching. CEM coarsens continuous covariates into bins using Sturges' rule and matches treated and control units exactly within bins, with region dummies matched exactly. CEM was attempted for all four agricultural comparisons and all four electoral comparisons for both the 1991 and 2023 elections. However, the resulting matched samples were substantially reduced for Two-way Exchange (direct exchange) and One-way Orthodox Departure, limiting statistical precision. Using the same 14 covariates, CEM matches 218 of the 863 Two-way Exchange treated neighborhoods to 771 control neighborhoods for the primary comparison. Post-matching balance is substantially improved: 18 of 19 covariates achieve $|\text{SMD}| < 0.1$, with only neighborhood area marginally unbalanced ($\text{SMD} = 0.15$). The key limitation relative to entropy balancing is that CEM discards approximately 75 percent of treated units through strict binning, restricting the estimating sample to the matched subsample.

5.3.2 Difference-In-Differences

I have used the DiD approach to complement my results of matching on agricultural production. Due to data limitations, DiD is only employed within agricultural analysis. The pre-treatment period consists of pooled observations from 1909, 1913, and 1914, while the post-treatment period consists of pooled Republican observations from 1936 to 1946. The cross-sectional analysis conducted with EB improves the

3. See Appendix Tables A.1 A.4 A.3 A.2.

balance on observed covariates, while DiD accounts for time-invariant differences between treated and comparison neighborhoods. The DiD analysis sample is restricted due to Ottoman province coverage, with modern districts, 88.2% of modern neighborhoods are covered. The DiD sample does not include Kars, Ardahan, and Artvin, controlled by the Russian Empire until 1917. 350 districts were covered out of 420 districts in 1935.

The DiD design relies on the "parallel trends assumption", which assumes that in the absence of the population exchange, treated and control units would have experienced similar changes in agricultural production.[4] This assumption cannot be verified directly, even though a pre-treatment differential trends test using only 1909, 1913, and 1914 is reported as an assessment. [5]

This design choice is directly informed by the placebo evidence discussed above: because cross-sectional matching alone does not equate treated and control neighborhoods on pre-treatment outcomes, difference-in-differences is treated as the primary source of causal evidence in this thesis, with the entropy balancing and coarsened exact matching results interpreted as descriptive and hypothesis-generating rather than independently causal.

The difference-in-differences specification is

$$\bar{y}_{it} = \alpha + \beta_1 \cdot \text{treated}_i + \beta_2 \cdot \text{post}_t + \beta_3 \cdot (\text{treated}_i \times \text{post}_t) + \boldsymbol{\gamma}' \mathbf{r}_i + \varepsilon_{it} \tag{5.1}$$

where $\bar{y}_{it}$ is mean log production of neighborhood i in period t (pre: 1909/1913/1914 pooled; post: 1936-1946), post_t is an indicator equal to one for the Republican period, and $\mathbf{r}_i$ is the vector of region dummies. The coefficient of interest is β_3, the treated $\times$ post interaction, which estimates the differential change between treated and comparison neighborhoods, net of time-invariant differences.

5.3.3 Placebo Check

To directly probe the credibility of the selection-on-observables assumption underlying the entropy balancing results, I apply the identical entropy balancing procedure

4. See Appendix Figure **??** – for the event study.

5. See Appendix Tables A.1 and A.2, and Figure A.1.

and covariate set used for the main post-treatment analysis to a genuinely pre-treatment outcome: mean log agricultural production during the Ottoman period (1909, 1913, 1914), using the same weights estimated for each comparison. If the matching covariates fully equated treated and control neighborhoods, this placebo regression should return coefficients statistically indistinguishable from zero, since no treatment effect can operate on an outcome measured before the treatment occurred.

This placebo check indicates that cross-sectional entropy balancing alone does not equate treated and control neighborhoods on outcomes that necessarily predate any causal effect of the exchange, and therefore cannot be treated as fully identifying a causal effect on its own. This finding directly motivates the difference-in-differences design presented above as the primary source of causal evidence in this thesis: because difference-in-differences nets out any time-invariant confounding, including the specific pre-existing production advantage this placebo check reveals, it relaxes the identifying assumption from full unconfoundedness on observables to the substantially weaker requirement that any remaining confounding be constant over time.

5.4 Agricultural Outcome Regression

The primary empirical analysis relies on a cross-sectional comparison of post-exchange outcomes across matched treatment and control neighborhoods. For each neighborhood, I average production outcomes (in tons) across all available years to obtain a single observation per neighborhood.

The primary specification uses entropy balancing weights:

$$\bar{y}_i = \alpha + \beta \cdot \text{treated}_i + \boldsymbol{\gamma}' \mathbf{r}_i + \varepsilon_i \tag{5.2}$$

where $\bar{y}_i$ is the mean log production of neighborhood i averaged across available years, treated_i is an indicator equal to one for treatment group neighborhoods, $\mathbf{r}_i$ is a vector of region dummies, and ε_i is an error term. Observations are weighted by the entropy balancing weights. The coefficient of interest β estimates the average treatment effect on the treated (ATT).

The robustness specification uses CEM subclass fixed effects:

$$\bar{y}_i = \alpha + \beta \cdot \text{treated}_i + \phi_s + \varepsilon_i \tag{5.3}$$

where ϕ_s are CEM subclass fixed effects that account for subclass-specific differences within matched groups, and observations are weighted by CEM weights. Standard errors are clustered at the neighborhood level in both specifications.

All production outcomes are transformed as $\ln(y_i + 1)$ before averaging. This transformation is motivated by the extreme right skew in raw production distributions and the prevalence of zero observations: 36 percent of neighborhood-year observations record zero olive production, 29 percent zero grape production, and 54 percent zero tobacco production. The log transformation compresses the distribution to approximate normality while preserving zero observations as $\ln(0+1) = 0$.

5.5 Electoral Outcome Regression

The same entropy balancing and CEM approaches are applied to neighborhood-level electoral outcomes from the 1991 parliamentary and 2023 general and presidential elections. Because electoral behavior may also reflect contemporary economic development rather than only the historical treatment, the electoral specification augments the baseline covariate set with two modern controls: log nighttime light intensity (`ln_nightlight`), extracted from the 2022 VIIRS annual composite as a proxy for local economic development, and log population density (`ln_pop_density`), computed as eligible voters per square kilometer, as a proxy for urbanization. The entropy balancing specification for electoral outcomes is therefore

$$y_i = \alpha + \beta \cdot \text{treated}_i + \boldsymbol{\gamma}'\mathbf{r}_i + \delta_1 \ln(\text{nightlight}_i) + \delta_2 \ln(\text{pop density}_i) + \varepsilon_i \tag{5.4}$$

where y_i is the relevant vote share or turnout rate in neighborhood i, observations are weighted by entropy balancing weights, and standard errors are clustered at the neighborhood level. Unlike the agricultural specification, electoral outcomes are not averaged across years, since each election is treated as a single cross-sectional observation. Including $\ln(\text{nightlight}_i)$ and $\ln(\text{pop density}_i)$ adjusts for any estimated electoral effect of the exchange is simply proxying for differences in contemporary

economic development between treatment and control neighborhoods, rather than reflecting a genuine historical legacy; if the coefficient β is robust to the inclusion of these controls, this concern is substantially mitigated.

5.6 Formal Mediation Analysis

To formally decompose the electoral effect of the exchange into a component operating through agricultural production and a component operating through other channels, this thesis employs the causal mediation framework of Imai et al. (2010). Formal mediation analysis, rather than an informal comparison of coefficients across specifications, is used because it provides a standardized decomposition (ACME, ADE, and proportion mediated) with an associated statistical framework for uncertainty quantification, following established practice in the political science literature on causal mediation. This approach fits two models: a mediator model,

$$\bar{y}_i^{\text{crop}} = \alpha_m + \beta_m \cdot \text{treated}_i + \boldsymbol{\gamma}_m' \mathbf{r}_i + \delta_m \ln(\text{nightlight}_i) + \delta_m' \ln(\text{pop density}_i) + \varepsilon_{mi} \tag{5.5}$$

and an outcome model,

$$y_i = \alpha_y + \beta_y \cdot \text{treated}_i + \lambda \cdot \bar{y}_i^{\text{crop}} + \boldsymbol{\gamma}_y' \mathbf{r}_i + \delta_y \ln(\text{nightlight}_i) + \delta_y' \ln(\text{pop density}_i) + \varepsilon_{yi} \tag{5.6}$$

both estimated by weighted least squares using entropy balancing weights. The Average Causal Mediation Effect (ACME), Average Direct Effect (ADE), and Total Effect are then computed from these two models following Imai et al. (2010), with confidence intervals obtained via quasi-Bayesian Monte Carlo simulations (1,000 simulations). The proportion mediated is the share of the total effect attributable to the ACME.

Mediation is estimated for the Two-way Exchange vs. Control comparison only, since this is the primary treatment comparison and the comparison for which the electoral effect is largest and most robustly estimated. Two mediators are treated as primary: olive and tobacco production, both of which exhibit treatment associations in the cross-sectional analysis that are broadly consistent with the difference-in-differences evidence presented in Chapter 6. Grape production is included as a deliberate contrast case: although grape shows a large and significant cross-sectional association with treatment, the difference-in-differences results indicate this association sub-

stantially predates the exchange (Table 6.5) and therefore suggest that much of the observed cross-sectional association predates the exchange. Any apparent mediation through grape production is consequently interpreted as reflecting this same pre-existing correlation rather than a genuine mediating pathway.

Mediation is estimated using Millet alliance vote share as the primary 2023 outcome, with Cumhur alliance vote share as a robustness check.

5.7 Limitations

Several limitations need to be addressed. Firstly, neighborhood-level production is not directly observed but distributed from the district total using extrapolated estimation on total agricultural land, which introduces measurement error at the neighborhood level. Secondly, the observed outcomes place my analysis approximately 15 years after the population exchange. Estimates capture the medium-run effects, rather than immediate post-exchange outcomes. Thirdly, the placebo check shows that entropy balancing does not fully equate treated and control neighborhoods even on pre-treatment outcomes, indicating that matching alone does not achieve full causal identification and motivating the difference-in-differences design as this thesis's primary causal specification. Fourthly, as with any observational design, unobserved confounding cannot be ruled out despite the matching procedures and geographic controls employed. Since post-treatment agricultural data is approximately 20 years after the exchange, I am unable to measure the immediate effect of the population exchange. Additionally, due to data limitations, I can only observe the electoral legacy of the immigrants 70 years after the treatment.

6. FINDINGS

6.1 Agricultural Analysis

As outlined in H1, consistent with the economic-disruption argument developed in Chapter 3 and settlement decisions disregarding prior agricultural expertise of the incoming refugees, treated neighborhoods would exhibit lower agricultural production than comparable control neighborhoods. Contrary to this expectation, Table 6.1 shows that the estimated coefficients are positive and statistically significant at the 99% confidence level for all four specified crops in direct exchange neighborhoods relative to the control group, the opposite sign from what H1 predicts. Likewise, grape and tobacco are also statistically significant for the One-way Orthodox Departure group at 99% confidence interval, compared with the control group. In contrast, olive production is not statistically distinguishable in Orthodox departed units, as seen in Table A.7. The same also holds for neighborhoods in which incoming Muslim refugees (*Lausanne exchangees*) are settled without prior Orthodox departure; however, coefficient magnitudes are smaller than those observed for Two-way Exchange, although both remain statistically significant. Despite a relatively small sample size, both estimates remain statistically significant at the 95% confidence interval. According to Table A.9, comparing direct exchange neighborhoods with neighborhoods that only experienced Orthodox departure reveals no statistical significance on any crops linked to Muslim refugee agricultural expertise. Overall, entropy balancing estimates indicate that any exchange-exposed neighborhoods exhibit higher production, except olive in non-direct-exchange neighborhoods, a pattern inconsistent with the directional expectation stated in H1. [1]

1. For Coarsened Exact Matching robustness results see Appendix Tables A.10, A.11. For Balkan refugee robustness analysis see Tables A.12, A.13. For unrestricted sample analysis (Mixed Neighborhoods) see Appendix Tables A.14, A.16, A.15, and A.17.

These findings are suggestive, but are unable to infer causality. Only post-treatment observations are not reliable to infer causal relationships; a DiD design is followed to provide complementary evidence on whether the observed cross-sectional associations persist once time-invariant differences are accounted for.

Table 6.1 H1: Effect of Population Exchange on Agricultural Production - Two-way Exchange vs. Control (Entropy Balancing)

Dependent Variable:	Logged total estimated production in tonnes		
Crop	Olive	Grape	Tobacco
Model:	(1)	(2)	(3)
Variables			
Two-way Exchange	0.2557***	0.5836***	0.3760***
	(0.0422)	(0.0549)	(0.0386)
Fit statistics			
N	50,409	50,409	50,409
R^2	0.27214	0.23738	0.18998

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable for each column is ln(crop production + 1), averaged across available years.

Table 6.2 Effect of Population Exchange on Olive Production, Restricted to Olive-Suitable Neighborhoods (suit_olive ≥ 2000)

Dependent Variable:	Logged total olive production in tonnes			
Comparison	Two-way Exchange vs Control	One-way Orthodox Departure vs Control	One-way Muslim Arrival vs Control	Two-way Exchange vs One-way Orthodox Departure
Model	(1)	(2)	(3)	(4)
Variables				
Treated	0.1759**	0.0161	-0.2659*	0.0968
	(0.0825)	(0.1307)	(0.1378)	(0.1442)
Fit statistics				
N	14,353	14,101	14,026	448
R^2	0.08990	0.15630	0.04489	0.32032

Clustered neighborhood-level standard errors in parentheses.
Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Notes: Sample restricted to neighborhoods with GAEZ olive suitability index. ≥ 2000.

According to Table 6.2 the Two-way Exchange olive effect remains directionally consistent and statistically significant across all three suitability-restricted samples, with the point estimate under the olive-suitability restriction (0.176) estimated less precisely, but not statistically distinguishable, from the full-sample estimate (0.256), consistent with the substantially reduced sample size (72 percent of the full sample excluded) rather than a genuine change in the underlying effect. The One-way

Orthodox Departure olive coefficient remains null across all four samples, reinforcing that a positive olive effect requires the joint presence of departure and settler arrival rather than either condition alone.

Table 6.3 Effect of Population Exchange on Grape Production, Restricted to Historically Grape-Producing Provinces

Dependent Variable:	Logged total grape production in tonnes			
Comparison	Two-way Exchange vs Control	One-way Orthodox Departure vs Control	One-way Muslim Arrival vs Control	Two-way Exchange vs One-way Orthodox Departure
Model:	(1)	(2)	(3)	(4)
Variables				
Treated	0.6180***	0.2408***	0.5403***	0.1479
	(0.0573)	(0.0716)	(0.2045)	(0.1288)
Fit statistics				
N	43,937	43,507	43,193	1,152
R^2	0.25198	0.22908	0.19949	0.32409

Neighborhood-clustered standard errors in parentheses.
Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Notes: Sample restricted to neighborhoods located in provinces that reported nonzero grape production in at least one Ottoman-era year (1909, 1913, or 1914). Because no dedicated GAEZ suitability layer exists for grape, this province-level revealed-cultivation restriction is used in its place. It is considerably weaker than the neighborhood-level GAEZ-based restrictions used for olive and tobacco (Tables 6.2 and 6.4), excluding only 11.5% of the full sample, compared with 72% and 62%, respectively. This difference reflects the substantially coarser nature of Ottoman province-level reporting relative to neighborhood-level agro-climatic suitability. The dependent variable in each column is $\ln(\text{grape production} + 1)$, averaged across available years. All comparisons are estimated via entropy balancing. All models include region fixed effects and neighborhood-clustered standard errors.

Across all three crops, the One-way Muslim Arrival coefficient is most stable, and in the case of olive and tobacco specifically strengthens, when the sample is restricted using that crop's own suitability criterion, but tends to lose statistical significance when restricted using a different crop's suitability index. This pattern is consistent with sample-size attrition on an agronomically irrelevant dimension reducing precision, rather than reflecting a genuine change in the underlying One-way Muslim Arrival effect; the crop-specific restrictions (Tables 6.2 for olive, 6.4 for tobacco) are accordingly the most informative test of each crop's robustness, while cross-crop restrictions serve primarily as a check on whether estimates survive a more conservative common-support requirement.

Table 6.4 Effect of Population Exchange on Tobacco Production, Restricted to Tobacco-Suitable Neighborhoods (`suit_tobacco` ≥ 2000)

Dependent Variable:	Logged tobacco production in tonnes			
Comparison	Two-way Exchange vs Control	One-way Orthodox Departure vs Control	One-way Muslim Arrival vs Control	Two-way Exchange vs One-way Orthodox Departure
Model:	(1)	(2)	(3)	(4)
Variables				
Treated	0.4237***	0.4480***	0.4594**	0.0729
	(0.0515)	(0.0903)	(0.1917)	(0.0976)
Fit statistics				
N	19,270	18,962	18,814	662
R^2	0.18450	0.10994	0.22422	0.29569

Clustered neighborhood-level standard errors in parentheses.
Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Notes: Sample restricted to neighborhoods with a GAEZ tobacco suitability index ≥ 2000. The dependent variable in each column is ln(tobacco production + 1), averaged across available years. Two-way Exchange, One-way Orthodox Departure, and One-way Muslim Arrival comparisons are estimated via entropy balancing. The Two-way Exchange versus One-way Orthodox Departure comparison is estimated via OLS with full covariate adjustment. All models include region fixed effects and neighborhood-clustered standard errors.

The Difference-in-Differences estimates provide complementary evidence regarding which cross-sectional associations remain after accounting for time-invariant differences. Comparing a reduced sample of neighborhoods, due to the coverage error of Ottoman-era records, suggests that some cross-sectional associations may partly reflect pre-existing differences between treated and comparison neighborhoods that remain after balancing observed covariates through entropy balancing. Firstly, Two-way Exchange neighborhoods see a roughly 68% increase in olive production relative to control, which is statistically significant at a 99% confidence interval. The positive DiD estimate for olive remains statistically significant in Two-way Exchange neighborhoods is consistent with a mechanism requiring both the departure of Orthodox producers and the arrival of incoming settlers. However, grape findings tell a different story from cross-sectional findings. The positive effect of the grape is dissolved in DiD; the estimated interaction is no longer statistically distinguishable from zero; the cross-sectional association appears largely attributable to pre-treatment differences. Table A.1 shows the pre-treatment average production for each crop, and direct exchange neighborhoods already have higher production than One-way Orthodox Departure and control as reported in Table 6.5. As a further check on the pooled specification, the treated $\times$ post interaction is also estimated at the annual level for the three crops with genuine year-by-year post-treatment observations (olive: 1936–1944; tobacco and wheat: 1944–1946), interacting treatment status with a full set of year dummies rather than a single pooled post indicator. Grape is excluded

from this specification, as its post-treatment source data reports a single pooled 1938–1942 average rather than distinct annual observations. The resulting event-study estimates, reported in Appendix **??**, are broadly consistent with the pooled results in Table 6.5: the olive effect for Two-way Exchange neighborhoods emerges immediately after 1923 and remains elevated across every subsequent observed year, and the divergence between Two-way Exchange and One-way Orthodox Departure for wheat, null for the former, consistently positive for the latter, is stable across the full post-treatment window rather than driven by any single year. Nevertheless, the effect for the grape is significant in Orthodox departure neighborhoods (One-way Orthodox Departure).

Secondly, the positive DiD estimates for tobacco remain statistically significant for both Two-way Exchange and One-way Orthodox Departure neighborhoods. Whereas the estimated coefficient is larger for One-way Orthodox Departure than for Two-way Exchange. On the other hand, the effect on wheat is similar to that of the grape; estimates treated by One-way Orthodox Departure neighborhoods are statistically significant. However, the effect is null for direct exchange neighborhoods (Two-way Exchange). The same pattern is also observed, according to Table A.1, wheat production in Two-way Exchange neighborhoods is higher than in One-way Orthodox Departure and Control in the pre-1923 period. The pre-treatment differences likely contribute to the cross-sectional estimates.

Table 6.5 H1: DiD, Ottoman vs. Republican Periods

Dependent Variable:	Logged production in tonnes							
Comparison	Two-way Exchange vs Control				One-way Orthodox Departure vs Control			
Crop	Olive	Grape	Tobacco	Wheat	Olive	Grape	Tobacco	Wheat
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables								
Treated	0.3514***	0.6762***	0.3347***	0.5921***	0.0730*	0.0320	0.2450***	0.0224
	(0.0399)	(0.0583)	(0.0276)	(0.0615)	(0.0398)	(0.0741)	(0.0415)	(0.0988)
Post	0.4365***	-0.0893***	0.5149***	-0.1140***	0.4389***	-0.0886***	0.5150***	-0.1129***
	(0.0094)	(0.0062)	(0.0071)	(0.0125)	(0.0094)	(0.0062)	(0.0071)	(0.0125)
Treated × Post	0.5170***	0.0239	0.2771***	0.0329	0.2679*	0.2039***	0.4145***	0.5955***
	(0.1023)	(0.0559)	(0.0508)	(0.0959)	(0.1467)	(0.0656)	(0.0820)	(0.1832)
Fit statistics								
N	58,388	86,487	59,447	59,787	57,768	85,682	58,725	59,208
R^2	0.28655	0.10111	0.22012	0.05103	0.27937	0.09827	0.21401	0.04951

Clustered (precinct) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable is mean ln(production + 1). Treated refers to the group indicated by the comparison label relative to group Control. Post is an indicator for the Republican (post-treatment) period. Region fixed effects included in all specifications; coefficients omitted for brevity. Clustered neighborhood standard errors in parentheses. Two-way Exchange vs Control = direct exchange vs. no exchange record. One-way Orthodox Departure vs Control = Orthodox departure only vs. no exchange record.

An important finding is that the effect of One-way Muslim Arrival neighborhoods either has null or negative statistically significant results across four crops, shown in Table 6.6. This is the only comparison consistent with the directional expectation of H1, in parallel with the importance of inherited agricultural capital. These findings are also robust for Balkan muhacir neighborhoods. [2] The suggestive results of the One-way Muslim Arrival sample, consistent with the hypothesis that inherited agricultural infrastructure contributes to subsequent production outcomes, are replicated in the same pattern for Balkan groups. Two-way Balkan refugee (Muslim Balkan refugee arrival, Orthodox departure) has a significantly positive effect only for tobacco and wheat, while the effect on olive and grape is null. However, One-way Balkan refugee (Muslim Balkan refugee arrival) results replicate the same pattern of One-way Muslim Arrival neighborhoods, supporting the potential channel of "arrival without inherited capital" may be associated with lower agricultural production. This implication is unlikely to be explained solely by the product of the small sample size of One-way Muslim Arrival neighborhoods, but negative results are robust in a completely different sample with similar historical trajectory.

The pattern of results across crops and comparison groups is consistent with a specific, theoretically grounded structure, rather than a uniform departure from H1. Three material trajectories are available to arriving settlers: economic disruption from skill mismatch, economic benefit from inherited productive capital, and destruction of that capital before settlers' arrival. The empirical pattern observed here suggests that which of these trajectories dominates depends jointly on (i) whether the crop in question requires long-lived, durable physical capital or can be readily re-established within a single growing season, and (ii) whether the treatment condition provides access to inherited capital at all.

For tree crops such as olive, mature groves represent capital accumulated over decades and cannot be rapidly replicated regardless of settler expertise. A positive effect for olive should therefore be observed only where both conditions for capital transfer are jointly satisfied: Orthodox departure, which releases the capital, and settler arrival, which provides someone to exploit it. This is precisely the pattern observed in Table 6.1 and confirmed causally in the difference-in-differences results: olive shows a significant positive effect for Two-way Exchange, but not for One-way Orthodox Departure alone. For annual or biennial crops such as tobacco and wheat, by contrast, production capacity is not embodied in long-lived physical assets in the same way, and can plausibly be re-established by any cultivator with access to the land, regardless of whether that cultivator inherited standing

2. See Appendix Table A.2.

infrastructure from a departing predecessor. Under this logic, a positive effect for these crops should be expected wherever land becomes newly available for cultivation, whether or not that land is subsequently settled by an arriving population, consistent with the observation that One-way Orthodox Departure alone produces significant positive difference-in-differences estimates for tobacco and wheat, of a magnitude comparable to or larger than Two-way Exchange. Grape falls between these two cases: as a perennial but comparatively shorter-lived crop than olive, and one whose cross-sectional advantage in Two-way Exchange neighborhoods is shown by the difference-in-differences results to substantially predate the exchange itself (Table 6.5), its apparent positive association is better explained by pre-existing regional viticultural geography than by any of the three material trajectories described above. Finally, One-way Muslim Arrival and its robustness counterpart, One-way Balkan Refugee, are the only conditions under which no inherited capital of any kind is available to arriving settlers. Under the disruption channel, this is precisely the configuration in which a negative production effect should be expected, and it is precisely the configuration in which negative and null estimates are in fact observed. Taken together, this pattern is consistent with a general principle: positive effects emerge specifically where productive capital, in whatever form the crop in question requires, is transferred from departing to remaining or arriving populations, while negative or null effects emerge where settlement occurs without such a transfer.

The general negative-effect prediction of H1 is not supported by these results. In contrast, this expectation appears to hold for arrival-without-departure cases, implying the absence of capital inheritance channel in One-way Muslim Arrival and One-way Balkan Refugee groups. Especially for olive, the capital-inheritance mechanism appear consistent with the production of tobacco, wheat, and grape increases across multiple treatment classifications. These results should not be interpreted as universally generalizable conclusions, nor do they allow for a simple confirmation or rejection of the hypothesis, but evidence consistent with multiple mechanisms operating across different settlement configurations.

The placebo results, reported in full in Appendix Tables A.3, A.6, A.5, and A.4. For Two-way Exchange versus Control, all four crops show large, statistically significant coefficients on the pre-treatment outcome (olive: 0.292, $p < 0.01$; grape: 0.667, $p < 0.01$; tobacco: 0.255, $p < 0.01$; wheat: 0.777, $p < 0.01$), indicating that Two-way Exchange neighborhoods were already agriculturally distinct from Control neighborhoods before 1923, in a manner the available covariates do not fully explain. The pattern is similar, though weaker for olive specifically, for One-way Orthodox Departure (olive null; grape, tobacco, and wheat significant) and for One-way Muslim

exchangee arrival (olive null; grape, tobacco, and wheat significant, with a caveat for the small underlying sample). The comparison between Two-way Exchange and One-way Orthodox Departure directly, by contrast, shows only a marginally significant placebo coefficient for olive and no significant divergence for the other three crops, suggesting these two groups were comparatively similar to one another on pre-treatment outcomes even where each differs substantially from Control.

Table 6.6 H1: DiD, Ottoman vs. Republican Periods - One-way Muslim Arrival vs Control

Dependent Variable:	Logged production in tonnes			
Comparison	One-way Muslim Arrival vs Control			
Crop	Olive	Grape	Tobacco	Wheat
Model:	(1)	(2)	(3)	(4)
Variables				
Treated	0.3520*	1.152***	0.4691***	1.112***
	(0.1906)	(0.2431)	(0.1062)	(0.2261)
Post	0.4395***	-0.0882***	0.5146***	-0.1124***
	(0.0094)	(0.0062)	(0.0071)	(0.0125)
Treated × Post	0.8309	-0.6345***	-0.3429*	-0.9443**
	(0.5446)	(0.2203)	(0.1881)	(0.4336)
Fit statistics				
N	57,392	85,118	58,279	58,831
R^2	0.27890	0.09755	0.21051	0.04968

Clustered (precinct) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable is mean ln(production + 1). Treated refers to the One-way Muslim Arrival group relative to Control. Post is an indicator for the Republican (post-treatment) period. Region fixed effects included; coefficients omitted for brevity. Clustered neighborhood standard errors in parentheses. One-way Muslim Arrival vs Control = exchangee arrival only, no prior Orthodox presence, vs. no exchange record.

6.2 Electoral Analysis

H2 hypothesis predicts differences in electoral behaviour consistent with greater support for opposition in neighborhoods exposed to Lausanne exchange refugees. Overall estimates, while accounting for socio-economic development and urbanization, indicate a consistent pattern.

Table 6.7 H2: 2023 Electoral Outcomes - Two-way Exchange vs. Control, (Entropy Balancing)

Dependent variable:	Vote share / turnout								
Outcome	AKP	CHP	Cumhur	Millet	Erdoğan R1	Kılıçdaroğlu R1	Erdoğan R2	Kılıçdaroğlu R2	Turnout
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Two-way Exchange	-0.0316***	0.0136**	-0.0383***	0.0333***	-0.0342***	0.0344***	-0.0356***	0.0356***	-0.0134***
	(0.0054)	(0.0058)	(0.0069)	(0.0067)	(0.0069)	(0.0070)	(0.0070)	(0.0070)	(0.0020)
Fit statistics									
N	48,702	48,702	48,702	48,702	48,702	48,702	48,702	48,702	48,702
R^2	0.1446	0.1603	0.2444	0.1856	0.2443	0.2255	0.2479	0.2479	0.0999

Clustered neighborhood standard errors in parentheses.
*Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Notes: R1 = first round; R2 = runoff. Treated = Two-way Exchange group relative to Control. Entropy balancing weights are used. Region fixed effects, log nightlight, and log population density are included. Standard errors are clustered at the neighborhood level.

According to Table 6.7, Two-way Exchange (*Lausanne exchangees*) neighborhoods show approximately 3 percentage points lower vote share for the governing party (AKP), the governing electoral alliance (Cumhur), and incumbent presidential candidate (Erdoğan) in both rounds of the presidential election in 2023. All estimated coefficients are statistically significant at conventional levels. Correspondingly, Two-way Exchange neighborhoods exhibit approximately 3 percentage points more for the opposition alliance (Millet) and opposition presidential candidate in both rounds. Additionally, these neighborhoods also show a 1.5 percentage points lower turnout rate. [3]

Table 6.8 H2: 2023 Electoral Outcomes, Restricted to Olive-Suitable Precincts - Two-way Exchange vs. Control (Entropy Balancing)

Dependent Variable:	Vote share / turnout								
Outcome	AKP	CHP	Cumhur	Millet	Erdoğan R1	Kılıçdaroğlu R1	Erdoğan R2	Kılıçdaroğlu R2	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Two-way Exchange	-0.0371***	0.0346***	-0.0534***	0.0502***	-0.0495***	0.0519***	-0.0514***	0.0514***	-0.0128***
	(0.0080)	(0.0086)	(0.0103)	(0.0104)	(0.0103)	(0.0107)	(0.0106)	(0.0106)	(0.0024)
Fit statistics									
N	13,728	13,728	13,728	13,728	13,728	13,728	13,728	13,728	13,728
R^2	0.11397	0.15994	0.19909	0.15769	0.19369	0.18657	0.19960	0.19960	0.03113

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Sample restricted to precincts with GAEZ olive suitability index ≥ 2000. EB weights, region fixed effects, clustered SEs at precinct level.

Similarly, One-way Muslim Arrival shows a larger coefficient with the same directional pattern, a higher vote share for the opposition party (CHP) and electoral

3. For all comparison group analyses, see Appendix Tables A.28, A.29, and A.30.

alliance (Millet), and a stronger lower vote share for the incumbent party (AKP) and incumbent electoral alliance (Cumhur). One-way Muslim Arrival neighborhood also shows a lower turnout rate. All results are statistically significant at the conventional levels. [4]

As seen in Table A.28 and A.30, the effect is generally smaller in magnitude, null for some outcomes, suggesting that the arrival component may play a larger role. Nevertheless, to note that One-way Orthodox Departure neighborhoods show a lower estimated vote share for AKP and Cumhur alliance and a lower turnout rate.

Table 6.9 H2: 2023 Electoral Outcomes, Restricted to Tobacco-Suitable Precincts - Two-way Exchange vs. Control (Entropy Balancing)

Dependent Variable:	Vote share / turnout								
Outcome	AKP	CHP	Cumhur	Millet	Erdoğan R1	Kılıçdaroğlu R1	Erdoğan R2	Kılıçdaroğlu R2	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Two-way Exchange	-0.0290***	0.0189***	-0.0400***	0.0373***	-0.0350***	0.0360***	-0.0358***	0.0358***	-0.0092***
	(0.0067)	(0.0072)	(0.0087)	(0.0087)	(0.0086)	(0.0089)	(0.0089)	(0.0089)	(0.0022)
Fit statistics									
N	18,719	18,719	18,719	18,719	18,719	18,719	18,719	18,719	18,719
R^2	0.11265	0.15590	0.20841	0.15925	0.20757	0.19915	0.21452	0.21452	0.03021

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Sample restricted to precincts with GAEZ tobacco suitability index $\geq$ 2000. EB weights, region fixed effects, clustered SEs at precinct level.

The estimated effect of Two-way Exchange status on 2023 electoral outcomes is robust to, and in several cases strengthens under, all three suitability-restricted specifications (Tables 6.8, 6.9, 6.10). Every coefficient remains statistically significant at conventional levels across all three restrictions, and the olive-suitability restriction in particular yields the largest estimated opposition-support effect of any specification in this thesis (a 5.34 percentage point reduction in Cumhur alliance vote share, compared to 3.83 percentage points in the unrestricted sample). This pattern indicates that the electoral finding is not an artifact of comparing agriculturally incomparable neighborhoods, and if anything strengthens once the sample is restricted to more agronomically homogeneous, and plausibly more broadly comparable, land. Given this thesis's mediation results indicate that agricultural production explains only a modest share of the electoral association, this strengthening under agricultural-suitability restriction should not be read as evidence that agricultural capital transfer specifically drives the larger effect; a more parsimonious interpretation is that these

4. For Coarsened Exact Matching robustness results see Tables A.34, A.35. For Balkan Refugee analyses see Tables A.36, and A.37. For full sample analyses (Mixed Neighborhoods) see Appendix Tables A.38, and A.39.

restrictions alter the comparison sample's regional composition in ways that happen to correlate with, but are not caused by, agricultural suitability itself. The one exception to this pattern of stability is CHP vote share under the grape-producing-province restriction, which weakens from significance at the 5 percent level in the full sample to marginal significance at the 10 percent level.[5]

Table 6.10 H2: 2023 Electoral Outcomes, Restricted to Historically Grape-Producing Provinces - Two-way Exchange vs. Control (Entropy Balancing)

Dependent Variable:	Vote share / turnout								
Outcome	AKP	CHP	Cumhur	Millet	Erdoğan R1	Kılıçdaroğlu R1	Erdoğan R2	Kılıçdaroğlu R2	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Two-way Exchange	-0.0318***	0.0106*	-0.0338***	0.0313***	-0.0303***	0.0304***	-0.0311***	0.0311***	-0.0120***
	(0.0055)	(0.0060)	(0.0070)	(0.0069)	(0.0070)	(0.0071)	(0.0071)	(0.0071)	(0.0022)
Fit statistics									
N	42,420	42,420	42,420	42,420	42,420	42,420	42,420	42,420	42,420
R^2	0.14678	0.16314	0.25398	0.19778	0.25047	0.23602	0.25469	0.25469	0.04763

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Sample restricted to precincts in provinces reporting nonzero Ottoman-era (1909, 1913, 1914) grape production. EB weights, region fixed effects, clustered SEs at precinct level.

The 1991 parliamentary elections are the earliest neighborhood-level election available. The overall pattern is broadly consistent with the 2023 findings despite a different party system. The Two-way Exchange neighborhoods show a lower vote share for right-wing DYP and RP, while exhibiting a 2 percentage point higher vote share for DSP. These estimates are statistically significant at conventional significance levels. Turnout rate in Two-way Exchange neighborhoods is in parallel with the 2023 elections, approximately 1 percentage point lower. Among the opposition parties, only DSP exhibits a statistically significant increase in vote share.

For Two-way Exchange neighborhoods, support for right-wing DYP and RP, and for One-way Muslim Arrival neighborhoods, support for the governing party is generally lower. The One-way Orthodox Departure neighborhoods show weaker and less consistent associations. The final comparison, Two-way Exchange vs. One-way Orthodox Departure are statistically indistinguishable from zero, which suggests limited additional electoral differences once Orthodox departure is held constant. [6]

5. For all group comparisons under restricted samples see Appendix for olive Tables A.19, A.18, A.20; for grape Tables A.26, A.25, and A.27; and for tobacco Tables A.22, A.23, and A.21.

6. For all comparison group analyses of 1991 elections, see Appendix Tables A.31, A.32, and A.33. For Coarsened Exact Marching robustness results see Tables A.40, A.41, A.42. For Balkan Refugee analyses see Tables A.49, and A.50. For full sample analyses (Mixed Neighbor-

Overall, despite substantial changes in parties and political competition, the findings are broadly consistent with a persistent opposition electoral orientation. The party-specific estimates additionally indicate that, in the 1991 context, this orientation was expressed primarily through greater support for DSP rather than uniformly across all opposition parties. This pattern is broadly consistent with the findings of Grigoriadis and Moschos (2023) for the Greek side of the exchange.

Turnout is consistently lower in treated neighborhoods in both the 1991 and 2023 elections. This pattern is consistent with persistent differences in political participation, even though the current analysis cannot distinguish among possible explanations. Several distinct mechanisms, not directly tested by this thesis, could plausibly generate the persistent opposition orientation and lower turnout documented above, and are worth distinguishing explicitly from the retrospective-voting and historical-persistence mechanism proposed in H2.

First, an *imported political culture* account would attribute the observed pattern not to a grievance response against the Turkish state, but to the political and religious dispositions that arriving Balkan Muslim populations already held prior to resettlement. Communities originating from more urbanized or Rumelian regions of the former Ottoman Balkans may have carried a comparatively more secular, Kemalist-aligned political orientation into their new neighborhoods independent of any disruption experienced during settlement itself, an account broadly consistent with the community-identity framing found in the Greek-side literature on refugee political integration (Kontogiorgi, 2006; Kritikos, 2005; Michalopoulos, 2017). Under this account, the observed effect reflects a durable cultural transmission from the origin community rather than a reaction to the disruption of resettlement per se.

Second, a *selective migration* account raises the possibility that the current residents of historically exchange-exposed neighborhoods are not, in fact, primarily the descendants of the original settlers. A century of subsequent internal migration could have altered the composition of these neighborhoods substantially, particularly given Turkey's rapid rural-to-urban transition over the twentieth century. If out-migration from, or in-migration into, these neighborhoods has been non-random with respect to political orientation, the estimated effect may in part reflect the political composition of who currently lives in these neighborhoods rather than a treatment effect transmitted within exchangee lineages.

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hoods), see Appendix Tables A.51, and A.52. For ideological preference analyses see Appendix Tables A.45, A.46, A.47, A.43, A.44, and A.48

Third, an *institutional and organizational* account would locate the mechanism not in individual political attitudes but in durable local institutions, religious institution density, civil society organizations, or local political networks and patronage structures, that developed differently in exchange-exposed neighborhoods and continue to shape political behavior independently of individual voters' beliefs or grievances. The neighborhood-level, cross-sectional and difference-in-differences designs employed in this thesis cannot adjudicate among these accounts, each of which implies a different underlying process despite generating an observationally similar aggregate pattern. Distinguishing them would require data this thesis does not have access to: measures of population turnover and migration history at the neighborhood level, survey or genealogical data capable of identifying exchangee descendants specifically within a modern population, or institutional data on local religious and civic organizational density. This thesis therefore documents the existence and persistence of the electoral pattern, and finds that agricultural production explains only a limited share of it (Section 6.3), without claiming to identify the specific mechanism through which the remainder operates. Lower turnout rate estimates provide complementary evidence on the long-run political legacy, which extends beyond party choice alone.

The consistency of the estimated direction across elections separated by more than three decades suggests that the observed electoral pattern is unlikely to be specific to a single election or party system.

Table 6.11 H2: Effect of Population Exchange on 1991 Party Vote Shares - Two-way Exchange vs. Control (Entropy Balancing)

Dependent Variable:	Vote share / turnout					
Party	DYP	ANAP	SHP	RP	DSP	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Two-way Exchange	-0.0107*	0.0086	-0.0027	-0.0162***	0.0223***	-0.0130***
	(0.0057)	(0.0058)	(0.0059)	(0.0048)	(0.0043)	(0.0031)
Fit statistics						
N	35,585	35,585	35,585	35,585	35,585	35,588
R^2	0.11964	0.02866	0.02767	0.19825	0.10931	0.27877

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable for each column is the party's vote share, or turnout, as indicated. Treated = Two-way Exchange relative to Control. Entropy Balancing weights, region fixed effects, log 1991 population density, clustered SEs at neighborhood level. ANAP was the incumbent governing party at the time of the 1991 election.

6.3 Mediation Analysis

H3 tests whether agricultural production statistically mediates the long-run electoral association. Total treatment effects remain statistically significant across all mediation specifications. The Average Direct Effect (ADE) accounts for the majority of the total effect, while the Average Causal Mediation Effect (ACME) is small to modest relative to the total effect. Agricultural production therefore explains only a limited share of the observed electoral association.

Olive is the most substantively plausible mediator among the three crops. Its ACME is statistically significant for both Millet and Cumhur alliance vote shares and accounts for approximately 9–10% of the total association. Although modest in magnitude, this pattern is consistent with partial mediation through olive production.

Tobacco also has a statistically significant ACME, but the indirect effect operates in the opposite direction from the total effect. Tobacco therefore functions as a small suppressing or inconsistent mediating pathway, offsetting approximately 3.2% of the total association rather than reinforcing it.

Grape exhibits the largest ACME among the three crops, accounting for approximately 13% of the total association. However, its Two-way Exchange treated-by-post coefficient is not statistically distinguishable from zero in the DiD analysis, while pre-treatment production levels indicate substantial pre-existing differences. Grape should therefore be interpreted as a contrast case rather than as a credible post-treatment mediating pathway.

Across all mediators, the direct effects remain considerably larger than the indirect effects. Most of the electoral association is not explained by the agricultural-production measures used here. The estimates therefore provide qualified evidence of partial mediation, strongest for olive, while tobacco operates as a small suppressing pathway and grape remains difficult to interpret causally.

The results are consistent with the existence of additional pathways beyond agricultural production. These may include community identity, intergenerational political socialization, or local social networks, although the present analysis cannot evaluate these mechanisms directly. Because formal mediation relies on stronger identifying assumptions than the primary regression analysis, these estimates should be interpreted as exploratory evidence regarding one candidate mediating pathway.

Table 6.12 H3: Causal Mediation Analysis of Agricultural Production and 2023 Electoral Alliance Vote Shares

Outcome	Mediator	ACME	95% CI	ADE	Total Effect	Prop. Mediated
Millet	Olive	0.003***	[0.003, 0.004]	0.031***	0.034***	9.5%
Millet	Tobacco	-0.001***	[-0.002, -0.001]	0.035***	0.034***	-3.2%
Millet	Grape	0.005***	[0.004, 0.005]	0.029***	0.034***	13.5%
Cumhur	Olive	-0.003***	[-0.004, -0.003]	-0.035***	-0.039***	9.0%
Cumhur	Tobacco	0.001***	[0.001, 0.002]	-0.040***	-0.039***	-3.2%
Cumhur	Grape	-0.005***	[-0.006, -0.004]	-0.033***	-0.038***	13.2%

Notes: ACME = Average Causal Mediation Effect; ADE = Average Direct Effect. Estimates are obtained using quasi-Bayesian Monte Carlo simulations (1,000 simulations) following Imai et al. (2010). Mediator and outcome models are estimated by weighted least squares using entropy-balancing weights. Both models include region fixed effects, log nightlight, and log population density as covariates. Grape is included as a contrast case. Unlike olive and tobacco, its effect on production is not supported by the difference-in-differences specification in Table 6.5. Its estimated ACME may therefore reflect the pre-existing cross-sectional association reported in Table A.1, rather than a causal mediating pathway.
$^{***}p < 0.01$.

7. CONCLUSION

In conclusion, the 1923 Lausanne Population Exchange provides a century-old, quasi-experimental setting to test whether forced migration affects economic and long-run political outcomes. This setting provides an opportunity to examine questions that are difficult to study using contemporary cases like Syria and Ukraine, while forced migration is a global phenomenon that is increasing at scale.

In light of the findings presented by the agricultural analysis, cross-sectional EB matching and DiD results show that direct exchange (Two-way Exchange) neighborhoods have higher olive and tobacco production. These findings do not support the directional expectation of H1. The DiD analysis provides supporting evidence in line with treatment effects for olive and tobacco production. By contrast, the positive cross-sectional association for grape production appears largely driven by pre-existing differences. The expected disruption is only observed in the One-way Muslim exchangee arrival neighborhoods, which is more consistent with a mechanism involving inherited agricultural capital (e.g., mature olive groves and vineyards) than with a settler skill-mismatch explanation. Overall, it seems that the economic "success" of Lausanne Exchangee neighborhoods in agricultural production is better explained by a different mechanism, most likely inherited capital and its transfer rather than human capital fit. This finding closely mirrors evidence from the receiving side of the exchange in Greece, where Kritikos (2005) and Kontogiorgi (2006) document that arriving Orthodox refugees were disproportionately settled on agricultural land expropriated from departing Ottoman Muslim populations, with almost none of this land redistributed to native Greek peasants instead. The present findings suggest that an analogous capital-transfer process operated on the Turkish side of the same exchange: arriving Balkan Muslim settlers benefited from productive agricultural capital, most plausibly mature olive groves, vacated by departing Orthodox producers, a pattern this thesis is, to the author's knowledge, the first to document quantitatively for the Turkish side.

The Two-way Exchange neighborhoods show statistically significantly greater support for opposition parties and alliances, together with lower support for governing parties and alliances. It is also important to note that the Two-way Exchange neighborhoods also portray a lower turnout rate. Notably, this pattern persists across the 1991 and 2023 elections. The persistence and direction of this electoral pattern situates the present findings within a growing quantitative literature on the political legacy of the exchange on its Greek side. Grigoriadis and Moschos (2023) document a persistent positive effect of Ottoman Greek refugee settlement on left-wing vote share across Greek provinces, an effect concentrated in areas of high Asia Minor refugee settlement but one that fades in the period following the restoration of democracy in 1974-75. Michalopoulos's (2017) work on Greek refugee neighborhoods similarly documents a shift toward Venizelist and subsequently Communist party support that persisted through the Second World War and beyond the 1974-75 democratic transition. The present findings extend this pattern to the Turkish side of the same exchange, documenting a directionally consistent opposition orientation across two electoral periods separated by more than three decades (1991 and 2023). Unlike the Greek-side finding of Grigoriadis and Moschos (2023), the present results show no evidence of the effect fading around a comparable political rupture on the Turkish side, though this thesis does not test for a specific structural break and cannot rule one out with the two electoral periods available. These findings provide support for H2 by documenting persistent differences in electoral behaviour across political contexts.

The formal mediation analysis identifies only a limited mediating role for agricultural production. Olive and tobacco, the two crops for which the Difference-in-Differences estimates show statistically significant treated-by-post coefficients, account for only a modest share of the observed electoral association. Olive exhibits a statistically significant indirect effect for both Millet and Cumhur alliance vote shares, explaining approximately 9–10% of the total association. Tobacco also has a statistically significant indirect effect, but it operates in the opposite direction from the total effect and therefore functions as a small suppressing pathway of approximately 3.2%. Grape produces the largest estimated ACME, accounting for roughly 13% of the association, but its DiD coefficient is not statistically distinguishable from zero and its production gap appears to reflect substantial pre-treatment differences. It should therefore be treated as a contrast case rather than as a credible post-treatment mediating pathway. Overall, agricultural production explains only a limited portion of the electoral association, while most of the estimated effect remains direct.

The mediation analysis indicates that agricultural production accounts for only a modest share of the total electoral association, a pattern broadly consistent with Murard's (2023) finding, using precinct-level data on the Greek side of the exchange, that Orthodox refugees were successfully integrated both economically and politically within a few generations. Notably, Murard (2023) attributes this integration in part to a concrete, identifiable channel: the construction of new schools for refugee communities shortly after their arrival, which closed the pre-existing literacy gap between refugee and native children, while related work on the same population documents complementary mechanisms operating through skill and know-how transfer (Murard & Sakalli, 2020). Read alongside the present results, this suggests that on the Turkish side of the same exchange, the observed electoral difference between historically exposed and comparable non-exposed neighborhoods is not primarily explained by the agricultural production channel examined here, though the Greek-side literature indicates that education and human-capital-related channels, distinct from the agricultural production channel tested in this thesis, are a plausible and previously identified alternative worth investigating directly in future work. The specific mechanism underlying the Turkish-side difference remains unidentified in this thesis.

Methodologically, this project is, to the author's knowledge, the first quantitative neighborhood-level analysis of the Turkish side of the population exchange. Moreover, the first study to jointly test agricultural and electoral outcomes with formal mediation analysis. This thesis also employs a DiD design to distinguish differential post-treatment changes from pre-existing cross-sectional differences.

This study has several limitations as well. The neighborhood-level production is estimated, not directly observed. There is approximately 15 years gap between treatment and the earliest post-treatment data for the agricultural analysis. Neither entropy balancing nor coarsened exact matching addresses bias arising from unobserved confounding. Rather, both procedures improve comparability between treated and comparison units with respect to observed covariates. Consequently, the validity of the resulting estimates continues to depend on the assumption that no important unobserved confounders remain after conditioning on the selected covariates. Furthermore, this gap extends to approximately 70 years for the earliest in 1991 in the electoral analysis. There are also potential unobserved confounders in both domains of outcome.

Future research should extend and enrich the mediating channels that would increase the explanatory power of the models to better understand the mechanisms underlying the electoral legacy. The main limitation of this project, which also points

to future improvements, would be to incorporate fine-grained neighborhood-level earlier electoral observations to complement this analysis. One important extension would be the extension of this project by comparing the findings systematically with the Greek-side estimates for exchange-wide symmetry testing. Lastly, local social mixing is a potential moderating factor and should be accounted for to observe whether it moderates the political effect.

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APPENDIX - DESCRIPTIVE STATISTICS

Table A.1 Descriptive Statistics by Treatment Group, Including Broader Muhacir Groups

Variable	Two-way Exchange	One-way Orthodox Departure	One-way Muslim Arrival	Broader Balkan Two-way Exchange	Broader Balkan One-way Arrival	Control
N	866	417	49	297	44	49,715
Area (km^2)	16.39 (15.47)	17.46 (17.70)	18.89 (19.47)	17.08 (16.20)	14.10 (9.14)	15.12 (22.38)
Coast dist. (km)	45.30 (53.80)	51.09 (56.20)	43.06 (51.76)	52.48 (53.76)	63.84 (63.18)	127.18 (108.16)
Port dist. (km)	71.11 (51.01)	76.22 (54.81)	73.22 (49.70)	81.63 (50.50)	100.23 (60.80)	153.30 (100.64)
Olive suit.	1613.95 (1517.17)	1079.83 (1360.84)	2064.38 (1452.69)	1478.01 (1537.78)	2003.57 (1605.97)	1288.43 (1824.98)
Tobacco suit.	2459.93 (1708.91)	1989.92 (1746.61)	3002.27 (1739.39)	2333.35 (1825.84)	2979.37 (1568.73)	1700.86 (1529.98)
Wheat suit.	4581.32 (2111.58)	3755.74 (2191.43)	5312.73 (2040.38)	4327.51 (2215.94)	5665.72 (1633.67)	4461.84 (1933.42)
Elevation (m)	563.94 (597.98)	806.47 (673.99)	318.92 (403.78)	679.27 (681.97)	323.77 (418.31)	880.84 (604.54)
Slope (°)	4.21 (3.15)	5.52 (3.59)	2.73 (2.39)	4.43 (3.44)	1.76 (1.52)	3.99 (3.12)
TRI	61.81 (46.35)	81.74 (51.34)	40.95 (40.08)	65.14 (50.48)	25.73 (23.26)	57.99 (45.39)
Temp. 1901–22 (°C)	11.45 (2.87)	9.84 (2.90)	12.60 (2.08)	10.80 (3.32)	12.67 (2.52)	11.19 (3.49)
Precip. 1901–22 (mm)	604.64 (92.79)	589.38 (79.35)	622.48 (95.57)	604.22 (81.30)	626.76 (102.38)	596.29 (129.24)
Railway dist. (km)	171.23 (143.12)	259.81 (138.15)	106.91 (104.44)	204.73 (147.71)	72.96 (86.89)	134.63 (128.02)
Istanbul dist. (km)	435.73 (298.14)	607.04 (317.20)	282.32 (221.25)	508.95 (328.98)	269.58 (206.80)	601.52 (339.31)
Ankara dist. (km)	410.90 (139.52)	446.73 (169.17)	409.95 (113.66)	454.79 (165.71)	407.83 (150.70)	426.53 (212.69)
Izmir dist. (km)	564.90 (352.49)	782.29 (353.32)	404.43 (250.34)	640.47 (389.61)	337.64 (238.77)	700.98 (391.26)
Major city dist. (km)	249.04 (180.23)	365.10 (212.78)	172.82 (95.16)	314.71 (213.52)	183.42 (119.14)	353.50 (247.26)
District center dist. (km)	14.51 (7.73)	15.28 (8.00)	13.80 (6.22)	15.38 (8.21)	15.90 (7.58)	16.33 (10.15)
Log nightlight, 2022	1.02 (0.91)	0.82 (0.70)	0.95 (0.70)	0.87 (0.82)	0.81 (0.59)	1.16 (1.23)
Log pop. density	3.63 (1.71)	3.29 (1.48)	3.39 (1.42)	3.39 (1.63)	3.29 (1.17)	3.82 (2.33)

Notes: Mean values with standard deviations in parentheses.

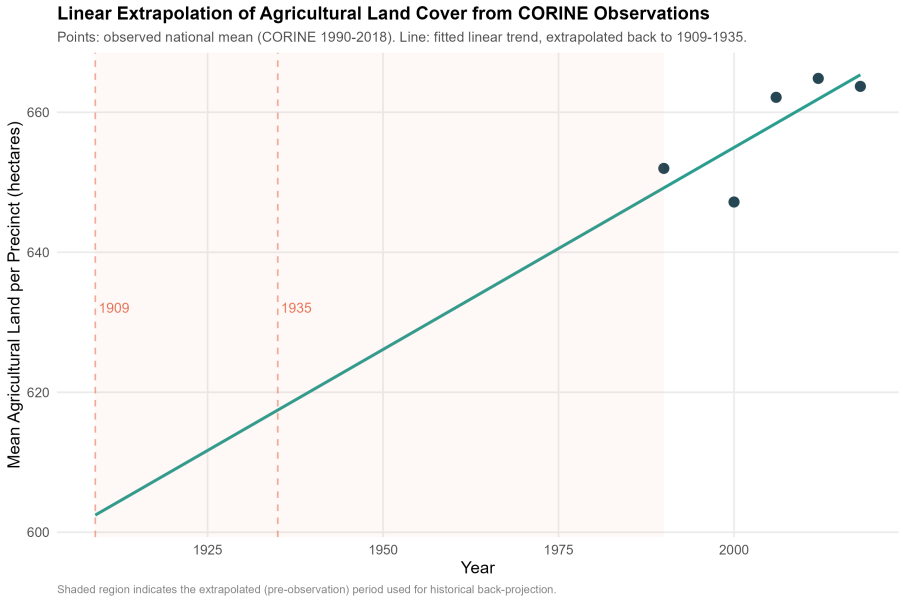


Figure A.1 Observed CORINE Land Cover: National Mean Agricultural Land per Precinct

Table A.2 Observed CORINE Land Cover: National Mean Agricultural Land per Precinct

Year	Mean Agricultural Land per Precinct (ha)
1990	651.98
2000	647.18
2006	662.13
2012	664.83
2018	663.70

Notes: Mean agricultural land area per precinct (hectares), averaged across all 51,297 precincts, at each observed CORINE Land Cover survey year. These values are the direct, unextrapolated observations used to fit the linear trend applied for historical back-projection (Figure A.1).

APPENDIX - DIFFERENCE-IN-DIFFERENCES

Table A.1 Pre-Trends Test: Treated × Year Interactions, 1909-1914 (Two-way Exchange vs. Control)

Dependent Variable:	Logged total estimated production in tonnes			
Crop	Olive	Grape	Tobacco	Wheat
Model:	(1)	(2)	(3)	(4)
Variables				
treated × production in 1909	-0.0407**	0.0480*	-0.0643**	-0.0285
	(0.0162)	(0.0290)	(0.0265)	(0.0319)
treated × production in 1913	0.0432*	0.0585**	-0.0938***	0.4014***
	(0.0238)	(0.0269)	(0.0123)	(0.0278)
Fit statistics				
N	133,446	133,446	119,946	125,756
R^2	0.27502	0.13472	0.11734	0.05362

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Reference year is 1914 (closest to treatment). A non-significant Treated × Year coefficient for 1909 and 1913 indicates no detectable divergence in production trends between treated and control neighborhoods prior to the 1923 exchange. Region fixed effects included; clustered SEs at neighborhood level.

Table A.2 Pre-Trends Test: Treated × Year Interactions, 1909-1914 (One-way Orthodox Departure vs. Control)

Dependent Variable:	Logged total estimated production in tonnes			
Crop	Olive	Grape	Tobacco	Wheat
Model:	(1)	(2)	(3)	(4)
Variables				
treated × production in 1909	-0.0572***	-0.0974**	-0.0157	-0.1858***
	(0.0146)	(0.0399)	(0.0331)	(0.0498)
treated × production in 1913	-0.0248	-0.0540	-0.2450***	0.2618***
	(0.0216)	(0.0383)	(0.0251)	(0.0502)
Fit statistics				
N	132,156	132,156	118,669	124,477
R^2	0.26960	0.13116	0.11141	0.05219

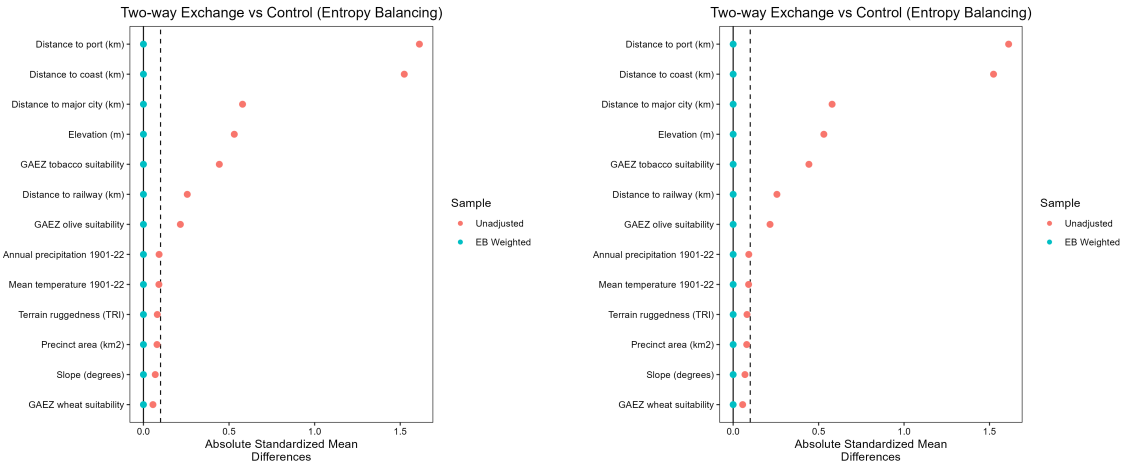
Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Reference year is 1914 (closest to treatment). Region fixed effects included; clustered SEs at neighborhood level.



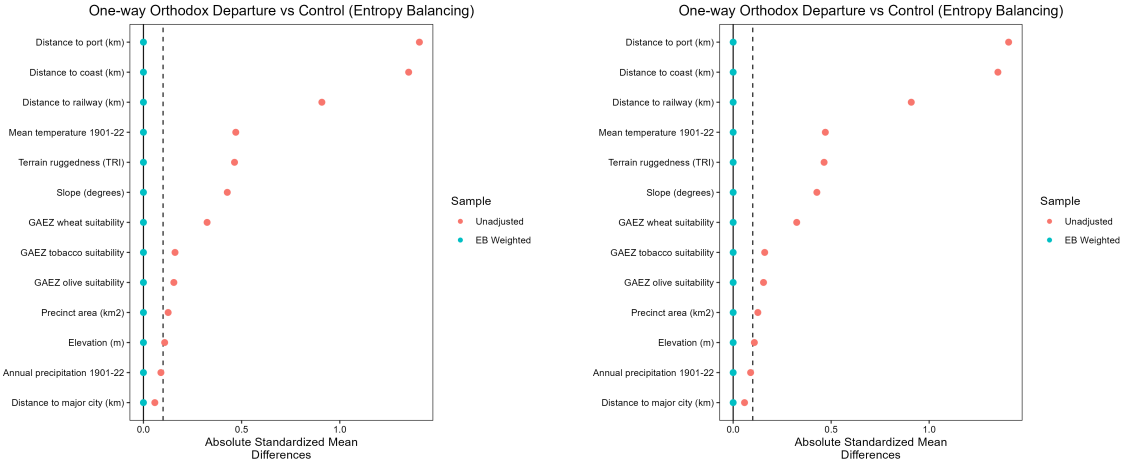
Figure A.1 Pre-treatment differential trends test

APPENDIX - BALANCE PLOTS AND ROBUSTNESS CHECKS



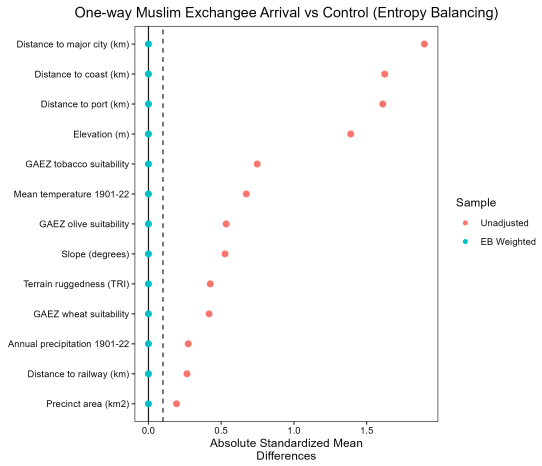
(a) Entropy Balancing (b) Coarsened Exact Matching

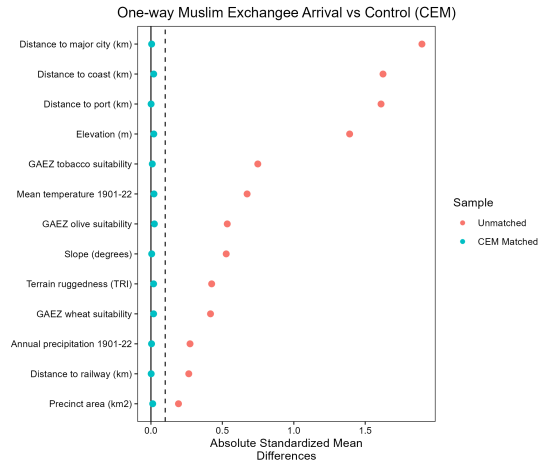
Figure A.1 Covariate Balance Before and After Matching, Two-way Exchange vs. Control comparison.



(a) Entropy Balancing (b) Coarsened Exact Matching

Figure A.2 Covariate Balance Before and After Matching, One-way Orthodox Departure vs. Control comparison.

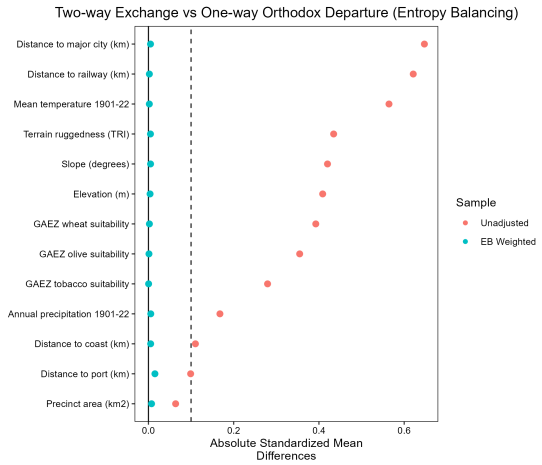


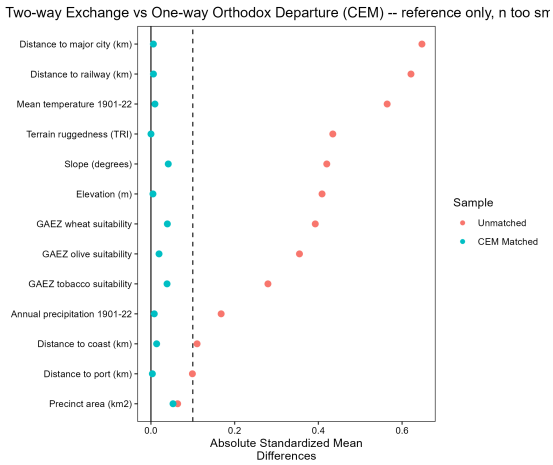


(a) Entropy Balancing (b) Coarsened Exact Matching

Figure A.3 Covariate Balance Before and After Matching, One-way Muslim exchangee arrival vs. Control comparison.

Difference-In-Differences





(a) Entropy Balancing (b) Coarsened Exact Matching

Figure A.4 Covariate Balance Before and After Matching, Two-way Exchange vs. One-way Orthodox Departure comparison.

Table A.1 Pre-Treatment Ottoman-Period Outcome Levels by Treatment Group

Crop	Two-way Exchange	One-way Orthodox Departure	One-way Muslim Exchangee Arrival	Control
Grape	2.13	1.30	2.75	1.58
Olive	0.82	0.31	1.04	0.34
Tobacco	0.74	0.68	0.86	0.29
Wheat	3.55	2.96	4.07	2.94

Notes: Mean ln(production + 1) in the Ottoman pre-treatment period (1909, 1913, 1914 pooled), by treatment group, for the sample used in the difference-in-differences analysis in Table 6.5. Two-way Exchange, One-way Muslim Exchangee Arrival, and, to a lesser extent, One-way Orthodox Departure already exhibit higher production levels than Control prior to the exchange for most crops, underscoring the importance of the difference-in-differences design in netting out pre-existing differences rather than relying on cross-sectional levels alone. With only two periods, pre and post, a formal test of parallel pre-trends is not possible; this table reports pre-treatment levels as the closest available diagnostic.

Table A.2 Robustness: DiD for Balkan Refugee Settlement

Dependent variable:	Mean ln(production + 1)							
Comparison	Two-way Balkan Refugee vs Control				One-way Balkan Refugee vs Control			
Crop	Olive	Grape	Tobacco	Wheat	Olive	Grape	Tobacco	Wheat
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables								
Treated × Post	0.1201	0.0348	0.3858***	0.7700***	-0.6101*	-0.5510**	0.2564	0.2213
	(0.1774)	(0.0817)	(0.0761)	(0.1812)	(0.3397)	(0.2145)	(0.1907)	(0.4226)
Fit statistics								
N	57,667	85,504	58,589	59,086	57,389	85,110	58,269	58,828
R^2	0.2791	0.0980	0.2124	0.0496	0.2784	0.0973	0.2106	0.0495

Clustered neighborhood standard errors in parentheses.
Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Notes: Treated refers to the group indicated by the comparison label relative to Control. Region fixed effects are included.

Placebo Checks

Table A.3 Placebo Check: Entropy Balancing Applied to Pre-Treatment (Ottoman, 1909-1914) Outcomes - Two-way Exchange vs. Control

Dependent Variable:	Logged total estimated production in tonnes			
Crop	Olive	Grape	Tobacco	Wheat
Model:	(1)	(2)	(3)	(4)
Variables				
Two-way Exchange	0.2922***	0.6669***	0.2552***	0.7765***
	(0.0382)	(0.0518)	(0.0274)	(0.0606)
Fit statistics				
N	44,482	44,482	44,482	44,482
R^2	0.35843	0.32998	0.14812	0.13079

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Dependent variable for each column is mean ln(pre-treatment production + 1) (Ottoman period, 1909–1914), using the identical entropy balancing weights estimated from the 14-covariate matching specification used for the main post-treatment analysis. A significant coefficient would indicate that matching on pre-determined covariates does not fully equate treated and control neighborhoods on a genuinely pre-treatment outcome, raising concern about the validity of the main cross-sectional comparison. Region fixed effects, clustered SEs at neighborhood level.

Table A.4 Placebo Check: Entropy Balancing Applied to Pre-Treatment (Ottoman, 1909-1914) Outcomes - One-way Orthodox Departure vs. Control

Dependent Variable:	Logged total estimated production in tonnes			
Crop	Olive	Grape	Tobacco	Wheat
Model:	(1)	(2)	(3)	(4)
Variables				
One-way Orthodox Departure	0.0461	0.2406***	0.1773***	0.4348***
	(0.0369)	(0.0664)	(0.0410)	(0.0966)
Fit statistics				
N	44,052	44,052	44,052	44,052
R^2	0.35179	0.30605	0.11708	0.11567

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Dependent variable for each column is mean ln(pre-treatment production + 1) (Ottoman period, 1909-1914), using the identical entropy balancing weights estimated for the main post-treatment analysis. Region fixed effects, clustered SEs at neighborhood level.

Table A.5 Placebo Check: Entropy Balancing Applied to Pre-Treatment (Ottoman, 1909-1914) Outcomes - One-way Muslim exchangee arrival vs. Control

Dependent Variable:	Logged total estimated production in tonnes			
Crop	Olive	Grape	Tobacco	Wheat
Model:	(1)	(2)	(3)	(4)
Variables				
One-way Muslim exchangee arrival	0.2637	0.8644***	0.3980***	1.098***
	(0.1850)	(0.2122)	(0.0995)	(0.2204)
Fit statistics				
N	43,738	43,738	43,738	43,738
R^2	0.22243	0.27279	0.22781	0.13136

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Dependent variable for each column is mean ln(pre-treatment production + 1) (Ottoman period, 1909-1914). Region fixed effects, clustered SEs at neighborhood level. Sample size is small (49 One-way Muslim exchangee arrival neighborhoods); the eastern-region dummy is dropped due to collinearity.

Table A.6 Placebo Check: Entropy Balancing Applied to Pre-Treatment (Ottoman, 1909-1914) Outcomes - Two-way Exchange vs. One-way Orthodox Departure

Dependent Variable:	Logged total estimated production in tonnes			
Crop	Olive	Grape	Tobacco	Wheat
Model:	(1)	(2)	(3)	(4)
Variables				
Two-way Exchange, vs. Orthodox Departure Only	0.1590*	0.2023	0.0458	0.1779
	(0.0926)	(0.1231)	(0.0565)	(0.1352)
Fit statistics				
N	1,152	1,152	1,152	1,152
R^2	0.39474	0.39765	0.14662	0.08303

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Dependent variable for each column is mean ln(pre-treatment production + 1) (Ottoman period, 1909-1914). Region fixed effects, clustered SEs at neighborhood level. The southeastern-region dummy is dropped due to collinearity.

Table A.7 H1: Effect of Population Exchange on Agricultural Production - One-way Orthodox Departure vs. Control (Entropy Balancing)

Dependent Variable:	Logged total estimated production in tonnes		
Crop	Olive	Grape	Tobacco
Model:	(1)	(2)	(3)
Variables			
One-way Orthodox Departure	0.0292	0.2107***	0.2593***
	(0.0362)	(0.0655)	(0.0532)
Fit statistics			
N	49,962	49,962	49,962
R^2	0.25357	0.20426	0.13217

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable for each column is ln(crop production + 1), averaged across available years. EB weights, region fixed effects, clustered SEs at neighborhood level.

Table A.8 H1: Effect of Population Exchange on Agricultural Production - One-way Muslim exchangee arrival vs. Control (Entropy Balancing)

Dependent Variable:	Logged total estimated production in tonnes		
Crop	Olive	Grape	Tobacco
Model:	(1)	(2)	(3)
Variables			
One-way Muslim exchangee arrival	0.2872	0.5185**	0.2645**
	(0.2300)	(0.2059)	(0.1310)
Fit statistics			
N	49,595	49,595	49,595
R^2	0.16309	0.18109	0.22596

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable for each column is ln(crop production + 1), averaged across available years. EB weights, region fixed effects, clustered SEs at neighborhood level.

Table A.9 H1: Effect of Population Exchange on Agricultural Production - Two-way Exchange vs. One-way Orthodox Departure (Entropy Balancing)

Dependent Variable:	Logged total estimated production in tonnes		
Crop	Olive	Grape	Tobacco
Model:	(1)	(2)	(3)
Variables			
Two-way Exchange, vs. Orthodox Departure Only	0.1214	0.1102	0.0640
	(0.0935)	(0.1226)	(0.0747)
Fit statistics			
N	1,279	1,279	1,279
R^2	0.34011	0.29783	0.24352

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable for each column is ln(crop production + 1), averaged across available years. EB weights, region fixed effects, clustered SEs at neighborhood level.

Table A.10 H1 Robustness: Effect of Population Exchange on Agricultural Production – Full Exchange and Orthodox Departure (CEM)

Dependent Variable:	Logged total estimated production in tonnes							
	Two-way Exchange				One-way Orthodox Departure			
Comparison	vs Control				vs Control			
Crop	Olive	Grape	Tobacco	Wheat	Olive	Grape	Tobacco	Wheat
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables								
Treated	0.1172*	0.2453***	0.0893	0.2837***	0.1745*	0.2963***	0.0778	0.0355
	(0.0628)	(0.0787)	(0.0589)	(0.0879)	(0.0928)	(0.1064)	(0.0829)	(0.1263)
Fixed effects								
Subclass	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics								
N	989	989	989	989	361	361	361	361
R^2	0.85191	0.86251	0.81042	0.82521	0.80420	0.87646	0.86594	0.84685

Neighborhood-clustered standard errors in parentheses.
Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Notes: The dependent variable for each column is listed in the header row. "Treated" refers to the group indicated by the comparison label (Two-way Exchange or One-way Orthodox Departure, as applicable) relative to the corresponding control group. Models use CEM weights, subclass fixed effects, and neighborhood-clustered standard errors. Each subclass s corresponds to a unique combination of coarsened covariate bins produced by the CEM algorithm. Only bins containing at least one treated and one control neighborhood are retained, and each neighborhood is assigned to exactly one subclass. Subclass fixed effects therefore restrict identification of β to within-subclass comparisons between treated and control neighborhoods sharing an identical coarsened covariate profile.

Table A.11 H1 Robustness: Effect of Population Exchange on Agricultural Production – Settler Arrival and Marginal Contribution (CEM)

Dependent Variable:	Logged total estimated production in tonnes							
Comparison	One-way Muslim Exchangee Arrival vs Control				Two-way Exchange vs One-way Orthodox Departure			
Crop	Olive	Grape	Tobacco	Wheat	Olive	Grape	Tobacco	Wheat
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables								
Treated	0.0217	0.4124	0.1260	0.3396	0.0063	-0.0785	-0.2704*	-0.0223
	(0.1197)	(0.3501)	(0.1707)	(0.2592)	(0.0053)	(0.1479)	(0.1591)	(0.0473)
Fixed effects								
Subclass	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics								
N	61	61	61	61	32	32	32	32
R^2	0.94330	0.92232	0.96496	0.91164	0.99984	0.96225	0.95909	0.99757

Neighborhood-clustered standard errors in parentheses.
Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Notes: The dependent variable for each column is listed in the header row. "Treated" refers to the group indicated by the comparison label (One-way Muslim Exchangee Arrival or Two-way Exchange, as applicable) relative to the corresponding comparison group. Models use CEM weights, subclass fixed effects, and neighborhood-clustered standard errors. Each subclass s corresponds to a unique combination of coarsened covariate bins produced by the CEM algorithm. Only bins containing at least one treated and one control neighborhood are retained, and each neighborhood is assigned to exactly one subclass. Subclass fixed effects therefore restrict identification of β to within-subclass comparisons between treated and control neighborhoods sharing an identical coarsened covariate profile.

Two-way Exchange vs. One-way Orthodox Departure cannot be replicated in CEM due to very small sample size.

Table A.12 Robustness: Effect of Balkan Refugee Settlement on Agricultural Production - Two-way Balkan Refugee vs. Control (Entropy Balancing)

Dependent Variable:	Logged total estimated production in tonnes		
Crop	Olive	Grape	Tobacco
Model:	(1)	(2)	(3)
Variables			
Two-way Balkan Refugee	0.0836	0.2882***	0.2308***
	(0.0618)	(0.0728)	(0.0569)
Fit statistics			
N	49,843	49,843	49,843
R^2	0.20590	0.23690	0.13525

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Dependent variable for each column is ln(crop production + 1), averaged across available years. Treatment group: Two-way Balkan Refugee, relative to Control. EB weights, region fixed effects, clustered SEs at neighborhood level. Treatment is reported for descriptive completeness and is not part of the primary Lausanne exchangee analysis.

Table A.13 Robustness: Effect of Broader Balkan Refugee Settlement on Agricultural Production - One-way Balkan Refugee Arrival vs. Control (Entropy Balancing)

Dependent Variable:	Logged total estimated production in tonnes		
Crop	Olive	Grape	Tobacco
Model:	(1)	(2)	(3)
Variables			
One-way Balkan Refugee Arrival	-0.0125	0.2533	0.3327**
	(0.1360)	(0.1918)	(0.1369)
Fit statistics			
N	49,590	49,590	49,590
R^2	0.17805	0.11316	0.17009

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Dependent variable for each column is ln(crop production + 1), averaged across available years. Treatment group One-way Balkan Refugee, relative to Control. EB weights, region fixed effects, clustered SEs at neighborhood level. Sample size is small (44 One-way Balkan Refugee neighborhoods); results should be interpreted with caution.

Robustness: Mixed Neighborhoods

The primary analysis excludes 82 neighborhoods (approximately 0.16 percent of the sample) classified as *mixed*: neighborhoods containing multiple settlements that individually fall into different treatment categories, most commonly a combination of Lausanne exchangee arrival (Two-way Exchange) alongside One-way Orthodox Departure recorded separately, or a combination of Lausanne exchangee and broader Balkan refugee arrival (Two-way Exchange and Two-way Balkan Refugee) within the same neighborhood. Because the agricultural and electoral outcome data are measured at the neighborhood level, a mixed neighborhood cannot be cleanly assigned to a single treatment category without an explicit resolution rule, and the primary specification avoids this ambiguity by dropping such neighborhoods entirely.

As a robustness check, I construct an alternative sample that reincorporates all 82 mixed neighborhoods under a single resolution rule, applied prior to and independently of any outcome analysis. The rule prioritizes categories as follows: if a neighborhood contains any Balkan refugee settlement, it is assigned to that category regardless of what else is present; failing that, if it contains any Lausanne exchangee

settlement (Two-way Exchange or One-way Muslim exchangee arrival), it is assigned accordingly; and only neighborhoods with no settler-arrival wave of any kind, but with an Orthodox departure recorded, are assigned to One-way Orthodox Departure. This rule is deliberately conservative with respect to the primary treatment definition: because the broader refugee category already serves as a secondary, non-legally-precise comparison group in this thesis, any neighborhood whose treatment status is genuinely ambiguous between a Lausanne exchangee wave and a broader Balkan Muslim migration wave is assigned to the broader category rather than the narrower one. This ensures that the primary Two-way Exchange and One-way Muslim exchangee arrival groups remain uncontaminated by settlements associated with the earlier, non-Lausanne-specific Balkan migration waves that began in 1877, at the cost of a small increase in heterogeneity within the already-secondary Two-way Balkan Refugee/One-way Balkan Refugee groups.

Table A.14 Robustness: Effect of Population Exchange on Agricultural Production, Mixed Neighborhoods Included - Two-way Exchange vs. Control (Entropy Balancing)

Dependent Variable:	Logged total estimated production in tonnes		
Crop	Olive	Grape	Tobacco
Model:	(1)	(2)	(3)
Variables			
Two-way Exchange	0.2449***	0.5652***	0.3911***
	(0.0409)	(0.0537)	(0.0387)
Fit statistics			
N	50,444	50,444	50,444
R^2	0.27726	0.23961	0.18702

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Dependent variable for each column is ln(crop production + 1), averaged across available years. Treatment group Two-way Exchange, relative to Control. Sample includes 82 previously excluded mixed neighborhoods, reassigned to a single treatment category using the priority-based resolution rule described in Appendix 5 (broader refugee presence takes priority in assignment, followed by Lausanne exchangee presence, with Orthodox departure only assigned where no settler-arrival wave is present). Group increases from 863 to 898 neighborhoods under this rule. EB weights, region fixed effects, clustered SEs at neighborhood level.

Table A.15 Robustness: Effect of Population Exchange on Agricultural Production, Mixed Neighborhoods Included - One-way Orthodox Departure vs. Control (Entropy Balancing)

Dependent Variable:	Logged total estimated production in tonnes		
Crop	Olive	Grape	Tobacco
Model:	(1)	(2)	(3)
Variables			
One-way Orthodox Departure	0.0380	0.2068***	0.2554***
	(0.0361)	(0.0649)	(0.0527)
Fit statistics			
N	49,968	49,968	49,968
R^2	0.27717	0.20177	0.13298

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Dependent variable for each column is ln(crop production + 1), averaged across available years. Treatment group One-way Orthodox Departure, relative to Control. Sample includes previously excluded mixed neighborhoods reassigned under the priority-based resolution rule; group increases from 416 to 422 neighborhoods. EB weights, region fixed effects, clustered SEs at neighborhood level.

Applying this rule reassigns 35 mixed neighborhoods to Two-way Exchange (increasing the group from 863 to 898 neighborhoods), 6 to One-way Orthodox Departure (416 to 422), and 41 to Two-way Balkan Refugee (297 to 338); no mixed neighborhood is assigned to One-way Muslim exchangee arrival under this rule, leaving that group unchanged at 49 neighborhoods. Table A.14 and Table A.15 report entropy balancing estimates for Two-way Exchange and One-way Orthodox Departure versus D on this expanded sample; Table A.16 reports the corresponding One-way Muslim exchangee arrival specification, which is numerically identical to the primary specification since no mixed neighborhoods were reassigned to that group. Table A.17 reports the Two-way Exchange vs. One-way Orthodox Departure comparison on this expanded sample.

Table A.16 Robustness: Effect of Population Exchange on Agricultural Production, Mixed neighborhoods Included - One-way Muslim exchangee arrival vs. Control (Entropy Balancing)

Dependent Variable:	Logged total estimated production in tonnes		
Crop	Olive	Grape	Tobacco
Model:	(1)	(2)	(3)
Variables			
One-way Muslim exchangee arrival	0.2872	0.5185**	0.2645**
	(0.2300)	(0.2059)	(0.1310)
Fit statistics			
N	49,595	49,595	49,595
R^2	0.16309	0.18109	0.22596

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Dependent variable for each column is ln(crop production + 1), averaged across available years. Treatment group One-way Muslim exchangee arrival, relative to Control. No mixed neighborhoods were reassigned to One-way Muslim exchangee arrival under the resolution rule; this specification is numerically identical to the primary clean-sample One-way Muslim exchangee arrival result. EB weights, region fixed effects, clustered SEs at neighborhood level.

Table A.17 Robustness: Effect of Population Exchange on Agricultural Production, Mixed neighborhoods Included - Two-way Exchange vs. One-way Orthodox Departure (Entropy Balancing)

Dependent Variable:	Logged total estimated production in tonnes		
Crop	Olive	Grape	Tobacco
Model:	(1)	(2)	(3)
Variables			
Two-way Exchange, vs. Departure Only	0.0870	0.1018	0.0677
	(0.0912)	(0.1195)	(0.0739)
Fit statistics			
N	1,320	1,320	1,320
R^2	0.35915	0.30004	0.23709

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Dependent variable for each column is ln(crop production + 1), averaged across available years. Comparison isolates the marginal contribution of Lausanne exchangee arrival, holding Orthodox departure constant, on the mixed-neighborhood-inclusive sample. EB weights, region fixed effects, clustered SEs at neighborhood level.

Electoral Analysis:

Table A.18 H2: 2023 Electoral Outcomes, Restricted to Olive-Suitable Precincts - One-way Orthodox Departure vs. Control (Entropy Balancing)

Dependent Variable:	Vote share / turnout								
Outcome	AKP	CHP	Cumhur	Millet	Erdoğan R1	Kılıçdaroğlu R1	Erdoğan R2	Kılıçdaroğlu R2	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
One-way Orthodox Departure	-0.0505***	0.0464***	-0.0674***	0.0554***	-0.0642***	0.0659***	-0.0706***	0.0706***	-0.0104
	(0.0125)	(0.0153)	(0.0168)	(0.0171)	(0.0169)	(0.0176)	(0.0177)	(0.0177)	(0.0072)
Fit statistics									
N	13,483	13,483	13,483	13,483	13,483	13,483	13,483	13,483	13,483
R^2	0.23587	0.26644	0.32828	0.26903	0.32928	0.32220	0.33023	0.33023	0.03454

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Sample restricted to precincts with GAEZ olive suitability index ≥ 2000. EB weights, region fixed effects, clustered SEs at precinct level.

Table A.19 H2: 2023 Electoral Outcomes, Restricted to Olive-Suitable Precincts - One-way Muslim Exchangee Arrival vs. Control (Entropy Balancing)

Dependent Variable:	Vote share / turnout								
Outcome	AKP	CHP	Cumhur	Millet	Erdoğan R1	Kılıçdaroğlu R1	Erdoğan R2	Kılıçdaroğlu R2	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
One-way Muslim exchangee arrival	-0.0535***	0.0359	-0.0668**	0.0548*	-0.0599**	0.0559*	-0.0620**	0.0620**	-0.0217**
	(0.0207)	(0.0294)	(0.0275)	(0.0303)	(0.0279)	(0.0310)	(0.0290)	(0.0290)	(0.0084)
Fit statistics									
N	13,408	13,408	13,408	13,408	13,408	13,408	13,408	13,408	13,408
R^2	0.18624	0.22305	0.34196	0.27411	0.30392	0.28494	0.30832	0.30832	0.05986

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Sample restricted to precincts with GAEZ olive suitability index ≥ 2000. EB weights, region fixed effects, clustered SEs at precinct level. The treated group is small (23 precincts); results should be interpreted with caution.

Table A.20 H2: 2023 Electoral Outcomes, Restricted to Olive-Suitable Precincts - Two-way Exchange vs. One-way Orthodox Departure (OLS with Covariate Adjustment)

Dependent Variable:	Vote share / turnout								
Outcome	AKP	CHP	Cumhur	Millet	Erdoğan R1	Kılıçdaroğlu R1	Erdoğan R2	Kılıçdaroğlu R2	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Two-way Exchange vs. One-way Orthodox Departure	0.0101	-0.0103	0.0141	-0.0019	0.0140	-0.0131	0.0195	-0.0195	-0.0003
	(0.0136)	(0.0162)	(0.0173)	(0.0177)	(0.0173)	(0.0180)	(0.0180)	(0.0180)	(0.0074)
Fit statistics									
N	441	441	441	441	441	441	441	441	441
R^2	0.39699	0.46047	0.51013	0.46682	0.50264	0.50704	0.51653	0.51653	0.09372

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Sample restricted to precincts with GAEZ olive suitability index ≥ 2000. Estimated via OLS with full covariate adjustment, as entropy balancing produced a degenerate solution (10 of 18 covariates unbalanced) in this restricted, doubly-similar-group comparison. Clustered SEs at precinct level.

Table A.21 H2: 2023 Electoral Outcomes, Restricted to Tobacco-Suitable Precincts - One-way Orthodox Departure vs. Control (Entropy Balancing)

Dependent Variable:	Vote share / turnout								
Outcome	AKP	CHP	Cumhur	Millet	Erdoğan R1	Kılıçdaroğlu R1	Erdoğan R2	Kılıçdaroğlu R2	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
One-way Orthodox Departure	-0.0357***	0.0266**	-0.0461***	0.0407***	-0.0442***	0.0411***	-0.0469***	0.0469***	-0.0073
	(0.0108)	(0.0112)	(0.0136)	(0.0132)	(0.0137)	(0.0137)	(0.0139)	(0.0139)	(0.0055)
Fit statistics									
N	18,419	18,419	18,419	18,419	18,419	18,419	18,419	18,419	18,419
R^2	0.15190	0.18187	0.22772	0.18494	0.22957	0.23076	0.24219	0.24219	0.04307

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Sample restricted to precincts with GAEZ tobacco suitability index ≥ 2000. EB weights, region fixed effects, clustered SEs at precinct level.

Table A.22 H2: 2023 Electoral Outcomes, Restricted to Tobacco-Suitable Precincts - One-way Muslim Exchangee Arrival vs. Control (Entropy Balancing)

Dependent Variable:	Vote share / turnout								
Outcome	AKP	CHP	Cumhur	Millet	Erdoğan R1	Kılıçdaroğlu R1	Erdoğan R2	Kılıçdaroğlu R2	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
One-way Muslim exchangee arrival	-0.0382*	0.0097	-0.0357	0.0284	-0.0296	0.0251	-0.0242	0.0242	-0.0110
	(0.0200)	(0.0260)	(0.0260)	(0.0281)	(0.0263)	(0.0286)	(0.0267)	(0.0267)	(0.0074)
Fit statistics									
N	18,271	18,271	18,271	18,271	18,271	18,271	18,271	18,271	18,271
R^2	0.22667	0.25376	0.36950	0.30445	0.34911	0.32942	0.35555	0.35555	0.01528

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Sample restricted to precincts with GAEZ tobacco suitability index ≥ 2000. EB weights, region fixed effects, clustered SEs at precinct level. The treated group is small (29 precincts); results should be interpreted with caution.

Table A.23 H2: 2023 Electoral Outcomes, Restricted to Tobacco-Suitable Precincts - Two-way Exchange vs. One-way Orthodox Departure (Entropy Balancing)

Dependent Variable:	Vote share / turnout								
Outcome	AKP	CHP	Cumhur	Millet	Erdoğan R1	Kılıçdaroğlu R1	Erdoğan R2	Kılıçdaroğlu R2	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Two-way Exchange . One-way Orthodox Departure	0.0115	-0.0263	0.0206	-0.0173	0.0210	-0.0228	0.0256	-0.0256	-0.0002
	(0.0137)	(0.0161)	(0.0182)	(0.0178)	(0.0182)	(0.0191)	(0.0190)	(0.0190)	(0.0047)
Fit statistics									
N	654	654	654	654	654	654	654	654	654
R^2	0.19514	0.24451	0.30462	0.24601	0.30175	0.29953	0.31110	0.31110	0.01106

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Sample restricted to precincts with GAEZ tobacco suitability index $\geq$ 2000. EB weights, region fixed effects, clustered SEs at precinct level.

Table A.24 H2: 2023 Electoral Outcomes, Restricted to Historically Grape-Producing Provinces - One-way Orthodox Departure vs. Control (Entropy Balancing)

Dependent Variable:	Vote share / turnout								
Outcome	AKP	CHP	Cumhur	Millet	Erdoğan R1	Kılıçdaroğlu R1	Erdoğan R2	Kılıçdaroğlu R2	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
One-way Orthodox Departure	-0.0188**	0.0018	-0.0184*	0.0127	-0.0167	0.0153	-0.0180	0.0180	-0.0050
	(0.0089)	(0.0096)	(0.0108)	(0.0106)	(0.0110)	(0.0110)	(0.0111)	(0.0111)	(0.0038)
Fit statistics									
N	42,003	42,003	42,003	42,003	42,003	42,003	42,003	42,003	42,003
R^2	0.11584	0.12052	0.19179	0.15436	0.18803	0.18203	0.19524	0.19524	0.05159

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Sample restricted to precincts in provinces reporting nonzero Ottoman-era grape production. EB weights, region fixed effects, clustered SEs at precinct level.

Table A.25 H2: 2023 Electoral Outcomes, Restricted to Historically Grape-Producing Provinces - One-way Orthodox Departure vs. Control (Entropy Balancing)

Dependent Variable:	Vote share / turnout								
Outcome	AKP	CHP	Cumhur	Millet	Erdoğan R1	Kılıçdaroğlu R1	Erdoğan R2	Kılıçdaroğlu R2	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
One-way Orthodox Departure	-0.0188**	0.0018	-0.0184*	0.0127	-0.0167	0.0153	-0.0180	0.0180	-0.0050
	(0.0089)	(0.0096)	(0.0108)	(0.0106)	(0.0110)	(0.0110)	(0.0111)	(0.0111)	(0.0038)
Fit statistics									
N	42,003	42,003	42,003	42,003	42,003	42,003	42,003	42,003	42,003
R^2	0.11584	0.12052	0.19179	0.15436	0.18803	0.18203	0.19524	0.19524	0.05159

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Sample restricted to precincts in provinces reporting nonzero Ottoman-era grape production. EB weights, region fixed effects, clustered SEs at precinct level.

Table A.26 H2: 2023 Electoral Outcomes, Restricted to Historically Grape-Producing Provinces - One-way Muslim Exchangee Arrival vs. Control (Entropy Balancing)

Dependent Variable:	Vote share / turnout								
Outcome	AKP	CHP	Cumhur	Millet	Erdoğan R1	Kılıçdaroğlu R1	Erdoğan R2	Kılıçdaroğlu R2	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
One-way Muslim exchangee arrival	-0.0505***	0.0342*	-0.0498**	0.0519**	-0.0540***	0.0496**	-0.0498**	0.0498**	-0.0137**
	(0.0159)	(0.0195)	(0.0193)	(0.0206)	(0.0198)	(0.0216)	(0.0207)	(0.0207)	(0.0057)
Fit statistics									
N	41,688	41,688	41,688	41,688	41,688	41,688	41,688	41,688	41,688
R^2	0.18817	0.24893	0.34433	0.29657	0.32329	0.30517	0.32583	0.32583	0.02834

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Sample restricted to precincts in provinces reporting nonzero Ottoman-era grape production. EB weights, region fixed effects, clustered SEs at precinct level. Sample size is small (46 precincts); results should be interpreted with caution.

Table A.27 H2: 2023 Electoral Outcomes, Restricted to Historically Grape-Producing Provinces - Two-way Exchange vs. One-way Orthodox Departure (Entropy Balancing)

Dependent Variable:	Vote share / turnout								
Outcome	AKP	CHP	Cumhur	Millet	Erdoğan R1	Kılıçdaroğlu R1	Erdoğan R2	Kılıçdaroğlu R2	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Two-way Exchange vs. One-way Orthodox Departure	-0.0106	-0.0007	-0.0056	0.0077	-0.0072	0.0052	-0.0055	0.0055	-0.0031
	(0.0108)	(0.0137)	(0.0151)	(0.0149)	(0.0152)	(0.0157)	(0.0162)	(0.0162)	(0.0039)
Fit statistics									
N	1,139	1,139	1,139	1,139	1,139	1,139	1,139	1,139	1,139
R^2	0.24162	0.26231	0.36272	0.30101	0.35472	0.34611	0.35890	0.35890	0.01852

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Sample restricted to precincts in provinces reporting nonzero Ottoman-era grape production. EB weights, region fixed effects, clustered SEs at precinct level.

Table A.28 H2: 2023 Electoral Outcomes - One-way Orthodox Departure vs. Control, Entropy Balancing

Dependent variable:	Vote share / turnout								
Outcome	AKP	CHP	Cum.	Mil.	Erd. R1	K.K. R1	Erd. R2	K.K. R2	Turn.
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Orthodox Departure	-0.0169**	-0.0012	-0.0192*	0.0060	-0.0153	0.0135	-0.0152	0.0152	-0.0103***
	(0.0082)	(0.0086)	(0.0100)	(0.0096)	(0.0101)	(0.0101)	(0.0101)	(0.0101)	(0.0034)
Fit statistics									
N	48,268	48,268	48,268	48,268	48,268	48,268	48,268	48,268	48,268
R^2	0.1137	0.1240	0.1838	0.1614	0.1823	0.1705	0.1890	0.1890	0.1429

Clustered neighborhood standard errors in parentheses.
*Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Notes: R1 = first round; R2 = runoff. Treated = One-way Orthodox Departure group relative to Control. Entropy balancing weights are used. Region fixed effects, log nightlight, and log population density are included. Standard errors are clustered at the neighborhood level.

Table A.29 H2: 2023 Electoral Outcomes - One-way Muslim exchangee arrival vs. Control, EB

Dependent variable:	Vote share / turnout								
Outcome	AKP	CHP	Cum.	Mil.	Erd. R1	K.K. R1	Erd. R2	K.K. R2	Turn.
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Mübadil Arrival	-0.0480***	0.0333*	-0.0481**	0.0511**	-0.0523***	0.0478**	-0.0491**	0.0491**	-0.0121**
	(0.0157)	(0.0192)	(0.0196)	(0.0200)	(0.0196)	(0.0215)	(0.0205)	(0.0205)	(0.0054)
Fit statistics									
N	47,900	47,900	47,900	47,900	47,900	47,900	47,900	47,900	47,900
R^2	0.1752	0.2431	0.3197	0.2880	0.3034	0.2851	0.3060	0.3060	0.0981

Clustered neighborhood standard errors in parentheses.
*Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Notes: R1 = first round; R2 = runoff. Treated = One-way Muslim exchangee arrival group relative to Control. Entropy balancing weights are used. Region fixed effects, log nightlight, and log population density are included. Standard errors are clustered at the neighborhood level. The One-way Muslim exchangee arrival sample is small; results should be interpreted with caution.

Table A.30 H2: 2023 Electoral Outcomes - Two-way Exchange vs. One-way Orthodox Departure, Entropy Balancing

Dependent variable:	Vote share / turnout								
Outcome	AKP	CHP	Cum.	Mil.	Erd. R1	K.K. R1	Erd. R2	K.K. R2	Turn.
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Direct Exchange	-0.0023	-0.0068	0.0006	0.0037	-0.0003	-0.0027	0.0008	-0.0008	-0.0021
	(0.0103)	(0.0132)	(0.0141)	(0.0143)	(0.0141)	(0.0145)	(0.0148)	(0.0148)	(0.0039)
Fit statistics									
N	1,266	1,266	1,266	1,266	1,266	1,266	1,266	1,266	1,266
R^2	0.2459	0.2715	0.3691	0.3076	0.3638	0.3514	0.3694	0.3694	0.1134

Clustered neighborhood standard errors in parentheses.
*Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Notes: R1 = first round; R2 = runoff. The comparison isolates the marginal contribution of Lausanne exchangee arrival, holding Orthodox departure constant. Entropy balancing weights are used. Region fixed effects, log nightlight, and log population density are included. Standard errors are clustered at the neighborhood level.

Table A.31 H2: Effect of Population Exchange on 1991 Party Vote Shares - One-way Orthodox Departure vs. Control (Entropy Balancing)

Dependent Variable:	Vote share / turnout					
Party	DYP	ANAP	SHP	RP	DSP	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
One-way Orthodox Departure	-0.0078	-0.0020	-0.0027	-0.0130*	0.0256***	-0.0040
	(0.0083)	(0.0086)	(0.0093)	(0.0078)	(0.0061)	(0.0045)
Fit statistics						
N	35,243	35,243	35,243	35,243	35,243	35,246
R^2	0.07231	0.04739	0.01407	0.15799	0.11606	0.26905

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable for each column is the party's vote share, or turnout, as indicated. Treated = One-way Orthodox Departure relative to Control. EB weights, region fixed effects, log 1991 population density, clustered SEs at neighborhood level.

Table A.32 H2: Effect of Population Exchange on 1991 Party Vote Shares - One-way Muslim exchangee arrival vs. Control (Entropy Balancing)

Dependent Variable:	Vote share / turnout					
Party	DYP	ANAP	SHP	RP	DSP	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
One-way Muslim exchangee arrival	0.0168	-0.0284**	0.0191	-0.0187*	0.0125	-0.0198**
	(0.0216)	(0.0143)	(0.0178)	(0.0110)	(0.0162)	(0.0079)
Fit statistics						
N	34,949	34,949	34,949	34,949	34,949	34,952
R^2	0.13140	0.04819	0.06848	0.36558	0.14517	0.20306

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable for each column is the party's vote share, or turnout, as indicated. Treated = One-way Muslim exchangee arrival relative to Control. Entropy Balancing weights, region fixed effects, log 1991 population density, clustered SEs at neighborhood level.

Table A.33 H2: Effect of Population Exchange on 1991 Party Vote Shares - Two-way Exchange vs. One-way Orthodox Departure (Entropy Balancing)

Dependent Variable:	Vote share / turnout					
Party	DYP	ANAP	SHP	RP	DSP	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Two-way Exchange, vs. Orthodox Departure Only	0.0068	0.0011	-0.0020	-0.0039	0.0001	-0.0048
	(0.0113)	(0.0104)	(0.0106)	(0.0088)	(0.0088)	(0.0058)
Fit statistics						
N	1,010	1,010	1,010	1,010	1,010	1,010
R^2	0.08658	0.04791	0.04135	0.31953	0.16507	0.27613

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: The dependent variable for each column is the party's vote share, or turnout, as indicated. Comparison isolates the marginal contribution of Lausanne exchangee arrival, holding Orthodox departure constant. EB weights, region fixed effects, log 1991 population density, clustered SEs at neighborhood level.

Note On Coarsened Exact Matching For Small-Sample Electoral Comparisons

Coarsened exact matching was attempted for all four electoral comparisons (Two-way Exchange vs. Control, One-way Orthodox Departure vs. Control, One-way Muslim exchangee arrival vs. Control, and Two-way Exchange vs. One-way Orthodox Departure) in both 1991 and 2023, following the same procedure used for the agricultural analysis. For three of these eight comparisons, the matched sample proved too small to support meaningful inference: One-way Muslim exchangee arrival vs. Control collapsed to a single matched pair in each treatment arm in 2023 (1 treated, 1 control) and to two matched pairs in 1991 (2 treated, 2 control), with substantial residual covariate imbalance in both years (6 of 22 and 3 of 21 covariates unbalanced, respectively). The 2023 Two-way Exchange vs. One-way Orthodox Departure comparison similarly collapsed to 2 treated and 2 control neighborhoods, with 3 of 22 covariates unbalanced.

Table A.34 H2 Robustness: 2023 Electoral Outcomes - Two-way Exchange vs. Control, CEM

Dependent variable:	Vote share / turnout								
Outcome	AKP	CHP	Cum.	Mil.	Erd. R1	K.K. R1	Erd. R2	K.K. R2	Turn.
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Direct Exchange	0.0276	-0.0180	0.0190	-0.0332	0.0092	-0.0115	0.0175	-0.0175	-0.0045
	(0.0322)	(0.0268)	(0.0394)	(0.0391)	(0.0404)	(0.0392)	(0.0382)	(0.0382)	(0.0084)
Fixed effects									
Subclass	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics									
N	78	78	78	78	78	78	78	78	78
R^2	0.7238	0.8057	0.7746	0.7657	0.7659	0.7625	0.7827	0.7827	0.6838

Clustered neighborhood standard errors in parentheses.
*Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Notes: R1 = first round; R2 = runoff. Treated = Two-way Exchange relative to Control. CEM weights and subclass fixed effects are used. Standard errors are clustered at the neighborhood level. Matched sample: 26 treated and 52 control neighborhoods.

Table A.35 H2 Robustness: 2023 Electoral Outcomes - One-way Orthodox Departure vs. Control, CEM

Dependent variable:	Vote share / turnout								
Outcome	AKP	CHP	Cum.	Mil.	Erd. R1	K.K. R1	Erd. R2	K.K. R2	Turn.
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Orthodox Departure	0.0275	0.0092	-0.0052	-0.0087	0.0066	0.0038	0.0011	-0.0011	0.0099
	(0.0389)	(0.0316)	(0.0358)	(0.0377)	(0.0387)	(0.0359)	(0.0420)	(0.0420)	(0.0165)
Fixed effects									
Subclass	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics									
N	24	24	24	24	24	24	24	24	24
R^2	0.8151	0.9063	0.9364	0.9029	0.9271	0.9364	0.9181	0.9181	0.8180

Clustered neighborhood standard errors in parentheses.
*Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Notes: R1 = first round; R2 = runoff. Treated = One-way Orthodox Departure group relative to Control. CEM weights and subclass fixed effects are used. Standard errors are clustered at the neighborhood level. Matched sample: 9 treated and 15 control neighborhoods; one of 22 covariates is marginally unbalanced.

This pattern mirrors the same limitation encountered in the agricultural analysis: CEM's requirement that treated and control units share an identical coarsened covariate profile across all matching dimensions becomes increasingly difficult to satisfy as the treated group shrinks, particularly for comparisons — such as One-way Muslim exchangee arrival, and Two-way Exchange vs. One-way Orthodox Departure — involving groups that are already small in the full sample (48–49 neighborhoods for One-way Muslim exchangee arrival; 850 and 416 neighborhoods for Two-way Exchange and One-way Orthodox Departure, respectively, but drawn from populations that are highly similar to one another on observable covariates). Because a matched sample of one or two pairs per arm cannot support reliable statistical

inference, CEM results are not reported for these three comparisons. Entropy balancing, which reweights the full control pool rather than requiring exact bin-level matches, remains the primary specification for all electoral comparisons throughout this thesis, and achieves full covariate balance (zero unbalanced covariates) in every case, including One-way Muslim exchangee arrival and Two-way Exchange vs. One-way Orthodox Departure. The one exception is the 1991 Two-way Exchange vs. One-way Orthodox Departure comparison, where CEM achieved an adequate matched sample (4 treated, 4 control, fully balanced) and is reported in Table A.42 despite its small size.

Table A.36 Robustness: 2023 Electoral Outcomes – Two-way Balkan Refugee vs. Control, EB

Dependent variable:	Vote share / turnout								
Outcome	AKP	CHP	Cum.	Mil.	Erd. R1	K.K. R1	Erd. R2	K.K. R2	Turn.
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Balkan refugee	-0.0081	-0.0161*	0.0009	-0.0068	0.0049	-0.0048	0.0040	-0.0040	-0.0085**
	(0.0092)	(0.0089)	(0.0112)	(0.0107)	(0.0110)	(0.0111)	(0.0111)	(0.0111)	(0.0035)
Fit statistics									
N	48,146	48,146	48,146	48,146	48,146	48,146	48,146	48,146	48,146
R^2	0.1197	0.1551	0.2048	0.1747	0.2102	0.1926	0.2155	0.2155	0.1681

Clustered neighborhood standard errors in parentheses.
*Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Notes: Balkan refugee arrival combined with Orthodox departure, relative to Control. Entropy balancing weights are used. Region fixed effects, log nightlight, and log population density are included. Standard errors are clustered at the neighborhood level. This comparison is not part of the primary Lausanne exchangee analysis.

Table A.37 Robustness: 2023 Electoral Outcomes – One-way Balkan Refugee vs. Control, EB

Dependent variable:	Vote share / turnout								
Outcome	AKP	CHP	Cum.	Mil.	Erd. R1	K.K. R1	Erd. R2	K.K. R2	Turn.
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Balkan refugee	0.0041	-0.0199	0.0168	-0.0119	0.0132	-0.0215	0.0121	-0.0121	-0.0094**
	(0.0207)	(0.0206)	(0.0276)	(0.0262)	(0.0270)	(0.0285)	(0.0284)	(0.0284)	(0.0047)
Fit statistics									
N	47,896	47,896	47,896	47,896	47,896	47,896	47,896	47,896	47,896
R^2	0.0656	0.0932	0.0722	0.1079	0.0696	0.0660	0.0654	0.0654	0.1183

Clustered neighborhood standard errors in parentheses.
*Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Notes: Balkan refugee arrival only, relative to Control. Entropy balancing weights are used. Region fixed effects, log nightlight, and log population density are included. Standard errors are clustered at the neighborhood level. This comparison is not part of the primary Lausanne exchangee analysis.

Table A.38 Robustness: 2023 Electoral Outcomes - Two-way Exchange vs. Control, Mixed Included

Dependent variable:	Vote share / turnout								
Outcome	AKP	CHP	Cum.	Mil.	Erd. R1	K.K. R1	Erd. R2	K.K. R2	Turn.
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Direct Exchange	-0.0293***	0.0103*	-0.0340***	0.0291***	-0.0302***	0.0303***	-0.0313***	0.0313***	-0.0138***
	(0.0053)	(0.0056)	(0.0067)	(0.0066)	(0.0067)	(0.0068)	(0.0068)	(0.0068)	(0.0020)
Fit statistics									
N	48,949	48,949	48,949	48,949	48,949	48,949	48,949	48,949	48,949
R^2	0.1491	0.1648	0.2505	0.1900	0.2505	0.2315	0.2541	0.2541	0.0984

Clustered neighborhood standard errors in parentheses.
*Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Notes: The sample includes 82 previously excluded mixed neighborhoods, reassigned under the priority-based resolution rule. Entropy balancing weights are used. Region fixed effects, log nightlight, and log population density are included. Standard errors are clustered at the neighborhood level.

Table A.39 Robustness: 2023 Electoral Outcomes - One-way Orthodox Departure vs. Control, Mixed Included

Dependent variable:	Vote share / turnout								
Outcome	AKP	CHP	Cum.	Mil.	Erd. R1	K.K. R1	Erd. R2	K.K. R2	Turn.
Model	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Variables									
Orthodox Departure	-0.0178**	0.0001	-0.0207**	0.0074	-0.0167*	0.0152	-0.0167*	0.0167*	-0.0102***
	(0.0082)	(0.0085)	(0.0099)	(0.0095)	(0.0100)	(0.0100)	(0.0100)	(0.0100)	(0.0034)
Fit statistics									
N	48,486	48,486	48,486	48,486	48,486	48,486	48,486	48,486	48,486
R^2	0.1175	0.1262	0.1872	0.1635	0.1858	0.1739	0.1926	0.1926	0.1424

Clustered neighborhood standard errors in parentheses.
*Signif. codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

Notes: The sample includes previously excluded mixed neighborhoods reassigned under the resolution rule. Entropy balancing weights are used. Region fixed effects, log nightlight, and log population density are included. Standard errors are clustered at the neighborhood level.

Left- Right Ideological Preference

As an additional robustness and comparative exercise, I also examine whether the ideological leftward pattern documented by Grigoriadis & Moschos (2023) for Greece extends to the Turkish case.

Table A.40 H2 Robustness: Effect of Population Exchange on 1991 Electoral Outcomes - Two-way Exchange vs. Control (CEM)

Dependent Variable:	Vote share / turnout		
Outcome	Left	Right	Turnout
Model:	(1)	(2)	(3)
Variables			
Two-way Exchange	-0.0030	0.0029	0.0155
	(0.0318)	(0.0316)	(0.0113)
Fixed-effects			
subclass	Yes	Yes	Yes
Fit statistics			
N	84	84	84
R^2	0.78332	0.78458	0.85178

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Dependent variable for each column is left vote share (SHP+DSP), right vote share (DYP+ANAP+RP), or turnout. Treated = Two-way Exchange relative to Control. CEM weights, subclass fixed effects, clustered SEs at neighborhood level. Matched sample: 33 treated, 51 control.

Table A.41 H2 Robustness: Effect of Population Exchange on 1991 Electoral Outcomes - One-way Orthodox Departure vs. Control (CEM)

Dependent Variable:	Vote share / turnout		
Outcome	Left	Right	Turnout
Model:	(1)	(2)	(3)
Variables			
One-way Orthodox Departure	0.0032	-0.0015	0.0108
	(0.0348)	(0.0339)	(0.0275)
Fixed-effects			
subclass	Yes	Yes	Yes
Fit statistics			
N	32	32	32
R^2	0.79569	0.80181	0.80685

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Dependent variable for each column is left vote share (SHP+DSP), right vote share (DYP+ANAP+RP), or turnout. Treated = One-way Orthodox Departure relative to Control. CEM weights, subclass fixed effects, clustered SEs at neighborhood level. Matched sample: 15 treated, 17 control; two of 21 covariates marginally unbalanced.

Table A.42 H2 Robustness: Effect of Population Exchange on 1991 Electoral Outcomes - Two-way Exchange vs. One-way Orthodox Departure (CEM)

Dependent Variable:	Vote Share / turnout		
Outcome	Left	Right	Turnout
Model:	(1)	(2)	(3)
Variables			
Two-way Exchange, vs. Orthodox Departure Only	-0.1212	0.1245	0.0062
	(0.0819)	(0.0817)	(0.0510)
Fixed-effects			
subclass	Yes	Yes	Yes
Fit statistics			
N	8	8	8
R^2	0.48473	0.49340	0.41997

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Dependent variable for each column is left vote share (SHP+DSP), right vote share (DYP+ANAP+RP), or turnout. Comparison isolates the marginal contribution of Lausanne exchangee arrival, holding Orthodox departure constant. CEM weights, subclass fixed effects, clustered SEs at neighborhood level. Matched sample: 4 treated, 4 control (fully balanced); given the small sample, results should be interpreted with caution.

Table A.43 Robustness: Effect of Balkan Refugee Settlement on 1991 Electoral Outcomes - Two-way Balkan Refugee vs. Control (Entropy Balancing)

Dependent Variable:	Vote Share		
Outcome	Left	Right	Turnout
Model:	(1)	(2)	(3)
Variables			
Two-way Balkan Refugee	-0.0032	0.0027	-0.0101**
	(0.0110)	(0.0111)	(0.0048)
Fit statistics			
N	35,147	35,147	35,150
R^2	0.05819	0.05685	0.34666

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Two-way Balkan Refugee relative to Control. EB weights, region fixed effects, log 1991 population density, clustered SEs. Not part of primary Lausanne exchangee analysis.

Table A.44 Robustness: Effect of Broader Balkan Refugee Settlement on 1991 Electoral Outcomes - One-way Balkan Refugee vs. Control (Entropy Balancing)

Dependent Variable:	Vote Share		
Outcome	Left	Right	Turnout
Model:	(1)	(2)	(3)
Variables			
One-way Balkan Refugee	-0.0325	0.0338	0.0039
	(0.0209)	(0.0213)	(0.0066)
Fit statistics			
N	34,947	34,947	34,950
R^2	0.02133	0.02174	0.17792

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: EB weights, region fixed effects, log 1991 population density, clustered SEs. Sample size is small; interpret with caution. Not part of primary Lausanne exchangee analysis.

Table A.45 H2: Effect of Population Exchange on 1991 Electoral Outcomes - Two-way Exchange vs. D (Entropy Balancing)

Dependent Variable:	Vote Share		
Outcome	Left	Right	Turnout
Model:	(1)	(2)	(3)
Variables			
Two-way Exchange	0.0197***	-0.0183**	-0.0130***
	(0.0072)	(0.0073)	(0.0031)
Fit statistics			
N	35,585	35,585	35,588
R^2	0.04238	0.04121	0.27877

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Dependent variable for each column is left vote share (SHP+DSP), right vote share (DYP+ANAP+RP), or turnout, as indicated. Treated = Two-way Exchange relative to Control. EB weights, region fixed effects, log 1991 population density (registered voters / neighborhood area), clustered SEs at neighborhood level. Nightlight is not included, as no satellite nightlight data exists prior to 1992.

Table A.46 H2: Effect of Population Exchange on 1991 Electoral Outcomes - One-way Orthodox Departure vs. Control (Entropy Balancing)

Dependent Variable:	Vote Share		
Outcome	Left	Right	Turnout
Model:	(1)	(2)	(3)
Variables			
One-way Orthodox Departure	0.0228**	-0.0228**	-0.0040
	(0.0108)	(0.0108)	(0.0045)
Fit statistics			
N	35,243	35,243	35,246
R^2	0.04822	0.04845	0.26905

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Dependent variable for each column is left vote share (SHP+DSP), right vote share (DYP+ANAP+RP), or turnout, as indicated. Treated = One-way Orthodox Departure relative to Control. EB weights, region fixed effects, log 1991 population density, clustered SEs at neighborhood level.

Table A.47 H2: Effect of Population Exchange on 1991 Electoral Outcomes - One-way Muslim exchangee arrival vs. Control (Entropy Balancing)

Dependent Variable:	Left-wing, right-wing vote share / turnout		
Outcome	Left	Right	Turnout
Model:	(1)	(2)	(3)
Variables			
One-way Muslim exchangee arrival	0.0315	-0.0303	-0.0198**
	(0.0233)	(0.0235)	(0.0079)
Fit statistics			
N	34,949	34,949	34,952
R^2	0.08192	0.08097	0.20306

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Dependent variable for each column is left vote share (SHP+DSP), right vote share (DYP+ANAP+RP), or turnout, as indicated. Treated = One-way Muslim exchangee arrival relative to Control. Entropy Balancing weights, region fixed effects, log 1991 population density, clustered SEs at neighborhood level. Sample size may be small given the 82.8% 1991 match rate; results should be interpreted with caution.

Table A.48 H2: Effect of Population Exchange on 1991 Electoral Outcomes - Two-way Exchange vs. One-way Orthodox Departure (Entropy Balancing)

Dependent Variable:	Left-wing, right-wing vote share / turnout		
Outcome	Left	Right	Turnout
Model:	(1)	(2)	(3)
Variables			
Two-way Exchange, vs. Orthodox Departure Only	-0.0018	0.0040	-0.0048
	(0.0139)	(0.0139)	(0.0058)
Fit statistics			
N	1,010	1,010	1,010
R^2	0.12916	0.12559	0.27613

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Dependent variable for each column is left vote share (SHP+DSP), right vote share (DYP+ANAP+RP), or turnout, as indicated. Comparison isolates the marginal contribution of Lausanne exchangee arrival, holding Orthodox departure constant. EB weights, region fixed effects, log 1991 population density, clustered SEs at neighborhood level.

1991 Elections Robustness Checks

Table A.49 Robustness: Effect of Broader Balkan refugee Settlement on 1991 Party Vote Shares - Two-way Balkan Refugee vs. Control (Entropy Balancing)

Dependent Variable:	Vote share / turnout					
Party	DYP	ANAP	SHP	RP	DSP	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Two-way Balkan Refugee	0.0033	0.0076	-0.0285***	-0.0082	0.0254***	-0.0101**
	(0.0096)	(0.0091)	(0.0074)	(0.0089)	(0.0073)	(0.0048)
Fit statistics						
N	35,147	35,147	35,147	35,147	35,147	35,150
R^2	0.06488	0.07249	0.04781	0.15359	0.12725	0.34666

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: EB weights, region fixed effects, log 1991 population density, clustered SEs.

Table A.50 Robustness: Effect of Broader Balkan Refugee Settlement on 1991 Party Vote Shares - One-way Balkan Refugee vs. Control (Entropy Balancing)

Dependent Variable:	Vote share / turnout					
Party	DYP	ANAP	SHP	RP	DSP	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
One-way Balkan Refugee	-0.0109	0.0579***	-0.0557***	-0.0132	0.0232	0.0039
	(0.0210)	(0.0152)	(0.0159)	(0.0160)	(0.0168)	(0.0066)
Fit statistics						
N	34,947	34,947	34,947	34,947	34,947	34,950
R^2	0.10989	0.08604	0.08899	0.12206	0.10287	0.17792

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: EB weights, region fixed effects, log 1991 population density, clustered SEs.

Table A.51 Robustness: Effect of Population Exchange on 1991 Party Vote Shares, Mixed neighborhoods Included - Two-way Exchange vs. Control (Entropy Balancing)

Dependent Variable:	Vote share / turnout					
Party	DYP	ANAP	SHP	RP	DSP	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Two-way Exchange	-0.0102*	0.0105*	-0.0053	-0.0153***	0.0217***	-0.0128***
	(0.0056)	(0.0057)	(0.0058)	(0.0048)	(0.0042)	(0.0030)
Fit statistics						
N	35,616	35,616	35,616	35,616	35,616	35,619
R^2	0.11794	0.02979	0.02839	0.19438	0.10601	0.27377

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Sample includes previously excluded mixed neighborhoods reassigned under the resolution rule. EB weights, region fixed effects, log 1991 population density, clustered SEs.

Table A.52 Robustness: Effect of Population Exchange on 1991 Party Vote Shares, Mixed neighborhoods Included - One-way Orthodox Departure vs. Control (Entropy Balancing)

Dependent Variable:	Vote share / turnout					
Party	DYP	ANAP	SHP	RP	DSP	Turnout
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
One-way Orthodox Departure	-0.0100	-0.0026	-0.0018	-0.0123	0.0268***	-0.0042
	(0.0082)	(0.0085)	(0.0091)	(0.0078)	(0.0061)	(0.0045)
Fit statistics						
N	35,248	35,248	35,248	35,248	35,248	35,251
R^2	0.06947	0.04728	0.01401	0.15932	0.11456	0.27037

Clustered (neighborhood) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Notes: Sample includes previously excluded mixed neighborhoods reassigned under the resolution rule. EB weights, region fixed effects, log 1991 population density, clustered SEs.